Structure, Properties and Performance of Solder Joints in Terrestrial vs. Reduced-Gravity Environments

NASA Grant Number 80NSSC23K0279

Year 1 Report

Principal Investigators

Siddhartha (Sid) Pathak & Ralph Napolitano

Department of Materials Science and Engineering, Iowa State University, Ames, Iowa, USA

NASA scientists Involved

Dr. Louise Strutzenberg

Dr. Binayak Panda

MSFC, National Aeronautics and Space Administration

Graduate Students

Manish Kumar [a]

Undergraduate Students

Soren Hellyer [a], Caleb Ogden [a]

Noah Madison [b], Joshua Kamp [b]

Department of Materials Science and Engineering, Iowa State University, Ames, Iowa, USA [a]

Department of Aerospace Engineering, Iowa State University, Ames, Iowa [b]

Research work at

IOWA STATE UNIVERSITY

Contents

Abstract: 4
Project Motivation: 6
Research Impact/Earth Benefits: 7
Background: 7
In-Space Soldering: 8
In-Space Soldering Investigation (ISSI 2003-04): 9
Research Plan: 11
Task A: 3-D Microstructural Characterization of Solder Samples at the Advanced Photon Source: 12
Task B: Sample Mounting and Mechanical Polishing: 13
Task C: Microstructural Characterization: 13
Lack of buoyancy force and natural convection causes increased porosity/voids in the microgravity solder: 13
Microstructure of microgravity vs. terrestrial solder: 14
Marangoni flow due to the large void and thermal convection causes accumulation of smaller voids and Pb-rich dendrites in the higher temperature region in the microgravity solder: 15
Task D: Characterizing the effects of the extreme elevated and cryogenic temperature on solder-joint properties: 17
Investigate the effect of aging on microstructure and properties of solders under ambient conditions: 17
Ex-situ nanoindentation to understand the effect of room temperature aging on near eutectic 40Pb-60Sn solder properties: 17
Ex-situ nanoindentation to understand the effect of room temperature aging on near eutectic 50Pb-50Sn solder properties: 18
Investigating the Effect of the cryogenic temperature on β-Sn to α-Sn phase transformation and resultant properties of the solder joints: 20
Future work: 22
Investigating the effect of accelerated aging on microstructure and properties of solders under elevated temperatures typically experienced by solder joints in space: 22
In-situ SEM micromechanical testing under extreme cryo temperature conditions: 22
Microstructural characterization of microgravity ISSI solders: 23
Awards 23
Conferences: 24

The Minerals, Metals, and Material Society 2023 Conference (TMS-2023) 24
International Space Station Research and Development conference 2023:..... 24
American Society of Gravitational and Space Research Conference 2023 24
Center for Advanced Non-Ferrous Structural Alloys (CANFSA) IRB Review meeting 2023 24
National Conference on Undergraduate Research (NCUR-2023) 24
The Symposium on Undergraduate Research & Creative Expression-2023 24
Conference presentation slides: 25
Please see the attached slides: 25
Attachment 1_ASGSR_Manish Kumar NASA Soldering_2023-11-16 25
Attachment 2_Hellyer_Ogden_ASGSR_SolderAging_Final 25
Monthly reports: 25
NASA PSI (**80NSSC23K0279**) January 2023 report 25
NASA PSI (**80NSSC23K0279**) February 2023 report 25
NASA PSI (**80NSSC23K0279**) March 2023 report 26
NASA PSI (**80NSSC23K0279**) April 2023 report 27
NASA PSI (**80NSSC23K0279**) June 2023 report 27
NASA PSI (**80NSSC23K0279**) July 2023 report 28
NASA PSI (**80NSSC23K0279**) September 2023 report 28
NASA PSI (80NSSC23K0279) October 2023 report 29
NASA PSI (**80NSSC23K0279**) November 2023 report 30
References: 31

Abstract:

The investigation proposed here combines experiments and modeling to elucidate the fundamental mechanisms, phenomenology, and process conditions that govern the integrity and performance of solder joints produced in terrestrial vs. reduced gravity environments, such as the microgravity conditions onboard the International Space Station (ISS). The technical research program plans to utilize solder samples from the In-Space Soldering Investigation (ISSI) experiments from the Physical Sciences Informatics (PSI) repository [1, 2], as well as expand into other non-ISSI solder compositions, and combine space- and ground-based experiments with advanced 3D materials characterization, micromechanical testing, and mesoscale modeling. In particular, the project addresses the formation and persistence of porosity through the reflow/filling/freezing processes and the deleterious effects on microstructure and mechanical properties of the solder joint. It has been established [1, 2] that porosity arising from flux volatilization, which is dispersed and expelled from the solder joint under terrestrial gravity, may become entrapped within the freezing solder material under microgravity conditions, given the absence of buoyancy-driven convection. Our overall goals are (i) to advance the current qualitative understanding of this phenomenon into the realm of alloy/process-specific quantitative description and prediction and (ii) to examine the effects of mechanically and acoustically stimulated flow while assessing their potential effectiveness as porosity mitigation strategies for solder-based fabrication processes in space. Considering a range of potential applications and materials, 3 solder alloys will be investigated, including the ISSI lead-based (Pb-Sn) solders, as well as lead-free (Sn-Ag-Cu and Sn-Au) solders, which have recently shown promise for high-performance joint applications due to their good thermal, electrical conductivities and excellent corrosion, fatigue resistance [3-5].

Additionally, solder joints in the space environment are subjected to large temperature variations such as – 120 ºC to +25 ºC for the Mars missions, and – 157 ºC to +121 ºC outside ISS. However, as deep space exploration missions become longer and more distant, the solder and other metal joints' current reliability is inadequate for the harsh conditions they encounter. Therefore, this project focuses on following objectives to address the solder joints' reliability in space: 1) Characterizing the effects of the lack of Earth's natural convective flow and buoyancy effects on solder-joint formation onboard ISS using the 40Pb-60Sn solders from In-Space Soldering Investigation (ISSI), 2) Examine the effects of mechanically and acoustically stimulated flow while assessing their potential effectiveness as porosity mitigation strategies for solder-based fabrication processes in space, and 3) Investigate the effects of elevated and cryo temperature conditions of space on solder properties.

In this work, using two microgravity solders from the ISSI study, one ISSI terrestrial solder along with a third set of freshly made terrestrial 40wt%Pb-60wt%Sn solder, we report results from the microstructural characterization and resultant micro-to-nano mechanical response of solders in terrestrial vs. microgravity environments, 1g vs. ~1×10^{-5}g under different temperature conditions. Results from the In-Space Soldering Investigation (ISSI) experiments performed aboard the International Space Station (ISS) have shown that soldering in microgravity is expected to be considerably different than their ground based counterparts due to Earth's natural convective flow and buoyancy effects being minimized in microgravity during melting and solidification. On Earth, tests showed the solder melting sagging, dropping, or running down the test wire due to gravity effect (Fig 1a). During this process, the entrapped gases and water vapor evolving from the circuit board itself are transported to the joint surface and mostly eliminated while the joint is molten.

In comparison, aboard the ISS the molten solder was observed to form an equilibrium "football" shape (Fig. 1b). The changes in geometric shape are believed to be due to the dominance of surface tension in microgravity[2, 6, 7]. Shortly after equilibrium, the accrued flux was seen to spin around the sample's surface. Liquid flux droplets rotating on sample surface gave indirect evidence of gravity independent thermocapillary flow, induced by internal temperature gradients within the molten solder ball. A flux bubble would migrate up a temperature gradient to the hottest spot, i.e. the components being joined. The trapped flux poses several concerns as it concentrates at sites like a repair joint, which can be detrimental to the desired strength as well as thermal and electrical conductivity from soldering. Also, the solder samples prepared aboard the ISS exhibited a considerable amount of internal porosity (Fig. 5) (about 13 times that of ISSI ground-based solders), as shown in the scanning electron microscope (SEM) micrographs in figure 6. The 3-D microstructural information from non-destructive micro-computed tomography (micro-CT) demonstrate the 3-D distribution of voids in the terrestrial solder (Fig. 4). The increased porosity is believed to be due to the lack of buoyant forces on flux and water vapor inherently present in the molten solder [1, 2, 6-8]. Entrapped gases will remain within the molten pool of solder until the solidification is complete, resulting in higher amount of voids, and compromising the mechanical integrity and electrical properties of the joint. Our current results also show a distinct difference of the distribution of the Sn-rich and Pb-rich, regions (Fig. 6) of the terrestrial vs. microgravity solders using electron-backscattered imaging.

A further consideration, not mentioned in the earlier ISSI reports[1, 2, 6-8] is the substantial effect of aging on the solder microstructure and properties over the past 17 years. Previous investigations have shown that Pb-Sn soldering alloys (due to their low melting temperatures) exhibit grain coarsening and reduced mechanical properties with age, even under ambient conditions (room temperature). Microstructural changes stem from the diffusion of Sn in to the Pb rich regions, creating a finer dispersion of grains as well as voids near phase boundaries. After aging 30 days in ambient conditions the microstructure and mechanical properties appeared to stabilize [9]. In earlier NASA PSI reports, we have discussed the aging properties of 40Pb-60Sn Solder. In the report, we demonstrate how the microstructure and properties of solder joints change as a function of aging in 50%Pb-50%Sn (Fig. 13).

Project Motivation:

Deployment of robust metal joining techniques in the space environment provide a solution towards sustainable, flexible missions through on-demand fabrication and repair of sustained damage to the spacecrafts, satellites, and habitat structures [10, 11]. However, long-duration space exploration missions in and beyond low earth orbit, such as Moon and Mars, present low gravity and an extreme temperature environment [12, 13] that requires novel metal joining solutions [13-15] to deploy and expand human exploration. Therefore, it is necessary to consider the new in-space metal joining (such as in-space soldering, among other techniques) approaches to electrical and structural component repair that reduces the need for large replacement components. Such in-space manual component repair will be essential for future long-duration human exploration missions beyond low Earth orbit [16-19]. However, previous investigations have shown that metal joining in space differs considerably from in terrestrial conditions.

Metal joining techniques in space are subjected to a) reduced gravity, b) reduced pressure, and c) a large temperature gradient than terrestrial metal joining. Additionally, the previous metal joining reports fall short of describing the fundamental structure-properties-processing relationships needed to develop robust in-space metal joining processes and process models. The persistent knowledge gap and the potential broad impacts associated with robust in-space metal joining capabilities clearly highlight the need for systematic investigation into the complex dynamics of melting, flow, volatilization, and solidification associated with metal joining in space. Therefore, this project's goal is to advance the current qualitative understanding of this phenomenon into the realm of solder alloy/process-specific quantitative description and prediction.

In addition, metal joints in the space environment are subjected to large temperature variations such as –120ºC to +25ºC for the Mars missions [20], – 230ºC to +120ºC for the missions to Moon, and – 157ºC to +121ºC outside ISS [21]. Similar to the human body system, the Mars exploration rovers cannot function well under extreme cold and hot temperatures. Therefore, to survive different phases of the mission, the rover's "vital organs," such as electronics/solder component, are situated inside a Warm Electronics Box (WEB) so that they do not get exposed to a temperature beyond ±40ºC [20]. To save electric power for the other function of the rover as well as to use electronics of low cost, a distributed, non-heated architecture is being pursued to place the electronics/solder components close to the system they interact with during future missions. However, as deep space exploration missions become longer and more distant, the solder and other metal joints' current reliability is inadequate for the harsh conditions they encounter [21, 22].

As part of the ongoing endeavors to establish an electronics and mechanical joint repair capability for extended space missions, developing advanced techniques and materials for soldering components on circuit boards and small structures in microgravity environments is essential. Addressing the reliability of solder joints under these multi-extreme environments is critical for the success of future deep space exploration missions. Specifically, the solder joints used in onboard electronics must be able to withstand cryogenic temperatures and large variations in temperature. For example, ISS makes 16 orbits of earth in 24 hours, traveling through 16 sunsets and sunrises. Due to this, anything outside ISS experiences extreme cryo and elevated temperature 16 times in a day. Therefore, in this project, we build upon previous in-space metal joining reports [23-26] to understand the effects of the lack of Earth's convective flow and buoyancy effects during the melting and solidification onboard space station along with studying the effect of temperature on solder properties. Considering a range of potential applications and materials, 3 solder alloys will be investigated, including near eutectic lead-based (40Pb-60Sn), Off-eutectic (50Pb-50Sn),

and lead-free (95.5Sn-3.8Ag-0.7Cu) solders, which have recently shown promise for high-performance joint applications due to their good thermal, electrical conductivities and excellent corrosion and fatigue resistance [3-5].

Research Impact/Earth Benefits:

The ongoing project is unique in that it combines both experimental and computational research on solder samples from the microgravity and ground based ISSI experiments, along with a comparative study on fresh solder samples. This project has the following research impact: (i) Our project objectives are a timely alignment with current NASA initiatives related to metals joining in reduced gravity environments. The three solder alloys for this work were chosen after extensive discussions with our NASA collaborators, and these alloy systems are currently being considered at NASA *as possible candidates for use in the NASA Mars Rover sample return missions*. (ii) We also anticipate that the results from this project will inform experiment design responsive to the (future) NASA proposal calls such as the NASA Artemis program, which seeks to land the first woman and next man on the Moon. (iii) The insights gained from this work can be extended to study effects of fluid interfaces with surface tension variations that are also common for many spaceflight-enabling technologies (e.g., boiling, heat transfer, combustion, welding, brazing, or soldering). We envision that this project will contribute towards advanced understanding of multiple processes that typically rely on gravity-driven separations in the terrestrial environment, such as the removal of gas bubbles from a liquid, which must be accomplished by other means in microgravity environments. (iv) the developed gravity-level-specific protocols (GLSPs) for several configurations can be used by future scientists and flight personnel for studying metal joining techniques such as welding, brazing, or soldering under reduced gravity.

Background:

Metal joining is a process that uses heat to melt or heat metal just below the melting temperature to combine two or more pieces of metal together to create a larger and more complex structure. These metal joining methods include welding, brazing, soldering, and adhesive bonding. Welding involves melting the metal at the joint to be joined, which then cools and solidifies, fusing the two pieces together. Brazing is similar to welding, but instead of melting the metal at the joint, a filler metal is used to join the pieces together. Soldering is similar to brazing, but the filler metal has a lower melting point and is used for joining smaller and more delicate components.

These metal joining processes such as in-space soldering are considered at NASA as possible techniques for the future long-duration missions such as *Moon's Artimis Program* and *NASA's Mars Rover Sample Return Mission* [27, 28]. NASA's Perseverance rover (currently seeking signs of ancient life in a large crater called Jezero on the Martian surface) is engaged in an audacious effort to bring samples of Martian rock and soil back to Earth. During the return voyage, the sample container will need to be packaged in a heavily protected capsule using metal joining processes – such as the in-space soldering processes (among other techniques) discussed in this proposal – in order to withstand re-entry into Earth's atmosphere.

Deploying these metal joining techniques in the space environment will reduce the human crew's dependency on the ground support system, such that crew can perform electronics and structural component repairs during the mission without logistical support from Earth but with minimal mass and volume. However, previous investigations have shown that metal joining in microgravity gravity environments differs considerably than terrestrial conditions [1, 10, 29-31]. The physics of terrestrial metal joining has been studied widely, and physics-based predictions of resulting metal

joint structure and properties are robust [10, 13]. The heat and mass transfer in terrestrial metal joining is greatly driven by gravity-induced natural convection. Key physical processes in terrestrial conditions include a) Heat transfer from the heat source to the metal alloy, b) Evaporation of fluxes, c) alloy melting, d) natural, Marangoni, electromagnetic convection, e) solidification, and f) defect formation such as porosity, solidification shrinkage, hot cracking, segregation. These physical processes further affect the structure and properties of the metal joint. However, these physical processes are altered in microgravity relative to the terrestrial conditions due to a) reduced gravity, b) reduced pressure, and c) large temperature gradient than the terrestrial metal joining [10, 13]. Therefore, expanding the understanding of the metal joining in space is needed to understand the critical changes in physical processes and generate data for calibration and validation of computational models.

In-Space Soldering: In soldering, an effective solder alloy will melt at a low temperature, readily wet the substrate components, and spontaneously flow into tight crevices under the action of capillary forces. In most commercial soldering processes, fluxes are necessary to suppress oxidation and to promote the desired wetting and spreading behavior, which is necessary for effective solder spreading and joint-gap filling during solder reflow. The non-destructive nature of the soldering process - with only short-time exposure to moderate temperatures and no melting or dissolution of the components themselves – makes this process widely applicable to a wide range of fabrication and repair scenarios, both manual and automated. Decades of NASA's mission success have been founded on the properties of Pb-Sn solders especially 40Pb-60Sn and 37Pb-63Sn. However, with increasing legislature against lead due to its toxic effect on human health, The United States Environmental Protection Agency (USEPA) also has lowered the Toxic Chemical Release reporting threshold for lead [32]. Due to rising concern over the use of lead in industry, Sn-Ag-Cu was invented [33, 34] by Dr. Anderson at the Ames Laboratory. It is a potential alternative to Pb-Sn solder for the Space applications due to good wettability, strength and thermal fatigue properties.

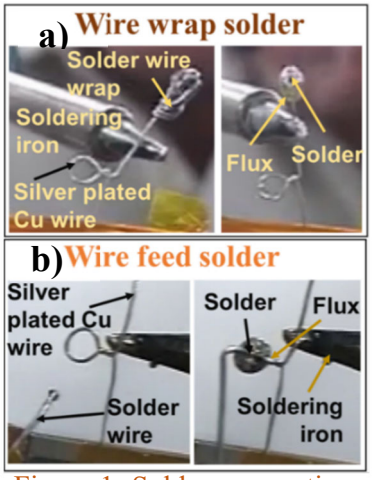


Figure 1: Solder preparation aboard the ISS a) Wire wrap solder b) Wire feed solder.

Indeed, soldering is integral to many industrial processes, and joints produced with lead-based and lead-free [33] solder alloys are ubiquitous in electrical, electronic, structural, and flow/pipe applications. As a flexible and robust joining process that can be applied across numerous materials systems, soldering is a potential solution to many *in-space* fabrication needs, including (i) in-flight repairs of electronic devices, plumbing systems, and structural components and (ii) sealing/resealing rigid containers with high structural/environmental integrity for long-term protective storage or transport. While the physical dynamics of soldering can be quite complex, an extensive and diverse experience base provides a sound framework for reliable process design in a typical terrestrial environment, under earth gravity with ambient temperature and pressure. The prospect of *in-space* processing to produce consistent solder joints under μg conditions, however, presents numerous challenges.

Unlike the broad ground-based experience with soldering processes, *space-based* experience with in-service components is limited to a small number of solder repairs on the Mir Space Station and the ISS, and reports of related inspections or other service characteristics are not available [35].

Likewise, reports of space-based experimental work is rather sparse, but some investigation has been done to assess the differences between the behavior of solders under microgravity conditions, as compared to those on Earth. Sounding rocket experiments, performed in the 1980s showed that capillary filling behavior for Pb-Sn solders was enhanced under reduced gravity, as compared to terrestrial conditions [36]. Subsequent Space Shuttle experiments, [37] further suggested that reduced buoyancy-driven convection may be responsible for entrapped flux in solder joints, an assertion supported by later parabolic flight experiments reported by Watson et al., where it was also shown that the type of flux and flux application procedure both influence the entrapment mechanism, where ground-based (low) porosity levels were most closely approached with a liquid flux that was heated and permitted to evaporate prior to solder wetting [35]. Experiments performed onboard the ISS [1, 2] revealed the dominant influence of thermocapillary flow in the absence of natural convection, further demonstrating the importance of joint configuration and process conditions in mitigation of the detrimental effects of solder joint porosity, such as the 32% decrease in joint shear strength associated with a 14% increase in internal porosity found by Thomas et al, comparing Sn-Ag-Cu joints produced in parabolic flights to those produced on the ground [38]. While a critical knowledge gap regarding the dynamics of pore formation and the influence of processing conditions persists, considerable efforts have been made toward mitigating the effects. For example, dispersion-strengthened or composite solders have been developed for enhanced strength, but these suffer from low ductility and do not offer the high fluidity and broad applicability to a range of joint configurations offered by monolithic alloy solders. Other strategies for reduction of porosity include induced flow by the action of a magnetic field on suspended magnetic (Fe or Fe-Co oxide) particles are currently being investigated [39].

The persistent knowledge gap and the potential broad impacts associated with robust in-space soldering capabilities clearly highlight the need for systematic investigation into the complex dynamics of melting, flow, volatilization, and solidification associated with soldering in space. The investigation proposed here aims to address this critical knowledge gap, advancing fundamental understanding and providing the quantitative relationships between solidification and flow characteristics, processing conditions, microstructure and porosity, and local/bulk mechanical properties for selected solder alloys. Therefore, this project builds upon the ISSI and utilizes samples from the ISSI experiments to enhance our understanding of soldering capabilities in microgravity. More details on ISSI experiments are given in the subsequent sections. In addition to applying novel materials characterization methods to *quantify* the deleterious effects of porosity and related microstructure on joint performance, we will extend previously reported efforts to minimize joint defects through rational design of the soldering parameters such as the substrate/solder composition and thermal profile in our new ISS Flight Opportunity experiments in Fall/Winter 2023.

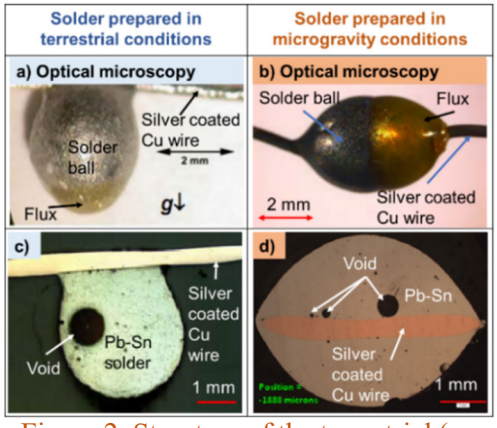


Figure 2: Structure of the terrestrial (a, c) vs. microgravity (b, d) solder using optical microscopy. Change in the geometric shape of the solder is due dominating surface tension in microgravity.

In-Space Soldering Investigation (ISSI 2003-04): This project utilizes the solder samples from the *ISSI experiments* performed in terrestrial and microgravity conditions (onboard ISS). ISSI experiments aboard the ISS present a unique opportunity to improve our fundamental understanding of the soldering phenomena

in a reduced gravity environment. The experimental procedure consisted of 86 soldering experiments conducted on the ISS in 2003-04, and an equivalent set of ground-based experiments using identical equipment and supplies. The ISS experiments were performed by taking pieces of silver-coated copper wire and bending and twisting them into multiple test coupon configurations. The selected lengths of 40wt%Pb-60wt%Sn wire were cut and 1) wrapped around the test coupons and melted using the soldering iron 2) fed on the similar test coupons that were heated by the soldering iron (Fig.1). The ground experiments involved conductively melting 10 cm lengths of 40Pb-60Sn (3.3 wt% rosin flux) solder onto a loop turned into the silver coated copper wire.

Results from the ISSI experiments performed aboard the ISS have shown that soldering in microgravity is expected to be considerably different than their ground-based counterparts due to Earth's natural convective flow and buoyancy effects being minimized in microgravity during melting and solidification [1, 40, 41]. On Earth, tests showed the solder melting, sagging, dropping, or running down the test wire due to the buoyancy (density) effect (Fig. 2a). In comparison, aboard the ISS, the molten solder was observed to form an equilibrium "football" shape (Fig. 2b). The changes in geometric shape are believed to be due to the dominance of surface tension in microgravity [2, 6, 7]. Shortly after equilibrium, the accrued flux was seen spinning around the sample's surface. As the solder started melting, the internal core of flux was released. When the solder was nearly melted, the released flux formed a drop and migrated to the bottom of the solder, which was closest to the heat source (soldering iron). Shortly after, the flux droplet started to spin around the molten solder (Fig. 3a-c). The rotation rate increased steadily and then reached a plateau (Fig. 3). This spinning continued until the heat source was removed, after which the flux drop slowed down and eventually stopped. The variation in flux rotation in three samples (Fig. 3) was probably because of different temperature gradients imposed by different soldering iron tip positions on Cu wire. The heat is initially transmitted through highly conductive Cu wire, which melts the solder. The solder edge closest to the heat source is the hottest, while the opposite edge is likely the coldest, which causes the temperature gradient across the solder surface. Therefore, flux rotation on the sample surface gave indirect evidence of gravity-independent thermocapillary flow induced by temperature gradients across the molten solder ball in μg conditions.

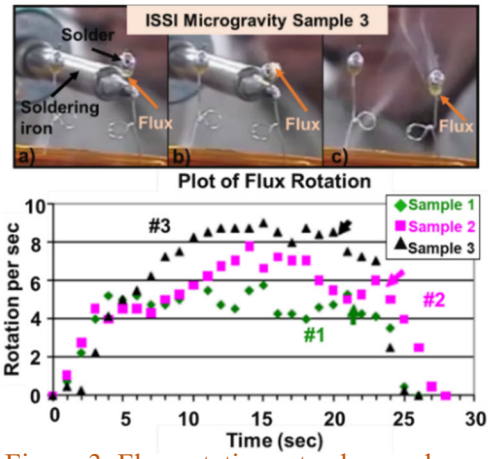


Figure 3: Flux rotation rate observed on three different microgravity wire wrap solders. Arrow indicates soldering iron removal from contact with Cu wire. Fig. (a-c) demonstrate the position of flux at different time due to rotation driven by thermocapillary flow.

Also, under typical soldering conditions, rapid reflow temperatures results in flux volatilization and bubble formation. In the presence of gravity, liquid solders tend to mix vigorously during the reflow period due to natural convection. This is beneficial, as any voids or bubbles that form due to evaporated fluxes are quickly swept to free surfaces and are thus removed from the solder joint. However, the situation is worse in the reduced gravity environment of space ($\sim 1\times10^{-5}$g) where, due to minimized buoyancy, bubbles have less chance to escape, thus complicating any in-space fabrication and repair operations. The ISSI data has demonstrated that in the *absence* of gravity, fluid motion is greatly reduced, driven mainly by solidification volume change, thermally induced density gradients, and Marangoni effects. Therefore, due to the lack of buoyancy force and slower

mixing in the absence of the natural convection, voids and bubbles are not swept away and become entrapped in the interior of the solder joint upon solidification in microgravity conditions. During the process of soldering, the liquid solder temperature near the substrate, which is in contact with the heater, is higher than that far away from the substrate/heater. Therefore, a distinct surface tension gradient will form with low surface tension in the high-temperature region and high surface tension in the low-temperature region. The effect of such a temperature gradient is predominant in the absence of the buoyancy force and natural convective flow [42]. Then the Marangoni force will tend to pull the voids towards the high-temperature regions such as the substrate.

The Marangoni force is: $F_M = 2K\pi \mid\sigma_T \mid T' R^2$ .....(1)

Where R = Void radius, σ_T = Temperature coefficient of surface tension, T' = Temperature gradient, K = empirical constant for the modification of departure from the linear theory of Young[43]

Total force on the growing void [42, 44]: $F_T = (F_B + F_P + F_I) - (F_D + F_M + F_S)$ .....(2)

F_B = Buoyancy, F_P = Pressure, F_I = inertia, forces F_D = Drag, F_M = Marangoni, F_S = Surface tension, forces

In the absence of the buoyancy in the microgravity, the dominating surface tension and Marangoni force drive the void detachment as well as void motion which causes the entrapped voids in the microgravity solder.

This entrapped porosity dramatically degrades the thermal and electrical properties of the solder and severely reduces the mechanical integrity of the joint. Such porosity is seen in conventional (Earth gravity) solder joints, but is much more prevalent in solder joints produced under reduced-gravity conditions (Fig. 2) [23-26]. As a result, any solder joints made under reduced gravity (such as in-orbit repair of electronic devices, tubing, and mechanical joints) are at risk of having their electrical or mechanical performance characteristics substantially degraded by entrapped porosity, increasing the manifold associated risks of subsequent device failures on astronauts and equipment in the unforgiving environment of space.

The ISSI experiments aboard the ISS present a unique opportunity to improve our understating of the effects of increased amount of porosity in the microgravity solder. Results from this study will enhance our fundamental understanding of the effects of surface tension driven convection phenomena during solidification processing operations such as brazing, soldering, and welding. Furthermore, the microgravity experiments represent a lowest gravity boundary condition. As such, these results could also be useful in predicting solidification behavior on other lower gravity environments (such as the moon or Mars).

Research Plan:

The ongoing project is being conduced under six major tasks. Each task is composed of integrated experimental (at Iowa State University, ISU, and Advanced Photon Source, APS) and modeling (at Iowa State University, ISU).

A breakdown of these three thrust areas is shown below in table 1.

Table 1: Tasks in which NASA PSI soldering project being conducted.

Task	Task Area	Year
Task A	3-D microstructural characterization	Year 1
Task B	Sample mounting and mechanical polishing	Year 1
Task C	Microstructural characterization	Year 1 and Year 2
Task D1	Micro-Mechanical Testing Strategies for Solders at Elevated and Cryo Temperatures	Year 1 and Year 2
Task D2	Aging effects under ambient temperatures	Year 1
Task F	Process Optimization and Active Mitigation of Porosity Defects	Year 2

Task A: 3-D Microstructural Characterization of Solder Samples at the Advanced Photon Source:

In order to characterize the 3-D distribution of Pb-rich dendrites and voids in the Pb-Sn solder, we performed non-destructive tomography at the advanced photon source. on the 0.7 mm x 1 mm x 2 mm sample. At APS, we focused on determining the feasibility of using the ISSI solder samples on APS beamlines. One particular challenge we encountered is the high attenuation of lead (Pb) in the Pb-Sn ISSI solders, which limits high-resolution tomography scans to sample thicknesses of 700 µm or less. Unfortunately, our current ISSI solder sample thicknesses (3-4 mm thick) are much larger than this limit. However, we prepared Pb-Sn solder specimens that were 4 mm x 1 mm in size. However, and we still had to cut them down to 700 µm x 1 mm rectangular sections in order to perform the tomography experiment, due to the resolution issues mentioned earlier. X-Ray tomography was conducted at 7-BM-B with a filtered white beam. The Pb-Sn solder specimens were scanned using a high-power X-ray head at 100kV source voltage and 102-amp source current. 2500 2-D projections will be obtained over a 180° scan range using a 0.056-degree rotation step and 0.5 degrees per sec rotation speed. Using these tomography parameters and solder sample thickness, we were able to achieve a resolution of 1 µm. A 3-D reconstruction

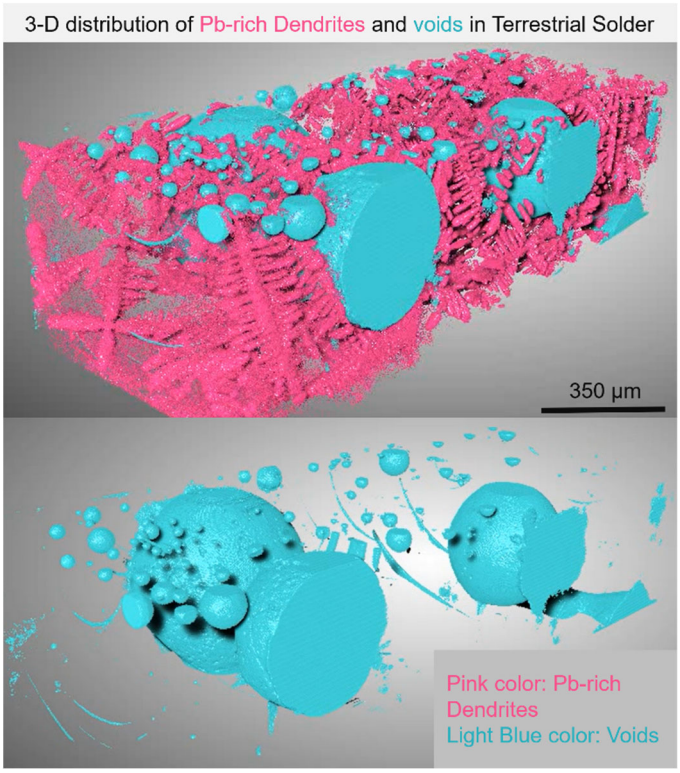


Figure 4. 3-D distribution of Pb-rich dendrites (pink color) and voids (light blue color) in terrestrial solder.

algorithm was used to reconstruct 2-D projections into a 3-D object (Fig. 4). Figure 35 demonstrates the distribution of Pb-rich dendrites and voids in the terrestrial solder.

Task B: Sample Mounting and Mechanical Polishing:

All solder specimens were prepared by embedding them in epoxy and then polishing the sur-face to a very fine finish. A very fine surface finish ensures that the samples are ready for both structural characterization as well as nano-mechanical testing, both of which are very sensitive to surface damage.

Task C: Microstructural Characterization:

Lack of buoyancy force and natural convection causes increased porosity/voids in the microgravity solder: To characterize the solder microstructure and void distribution, serial sectioning was carried out. The analysis conducted on the distribution of voids in the microgravity and terrestrial samples revealed a significant disparity in the void content between the solder samples.

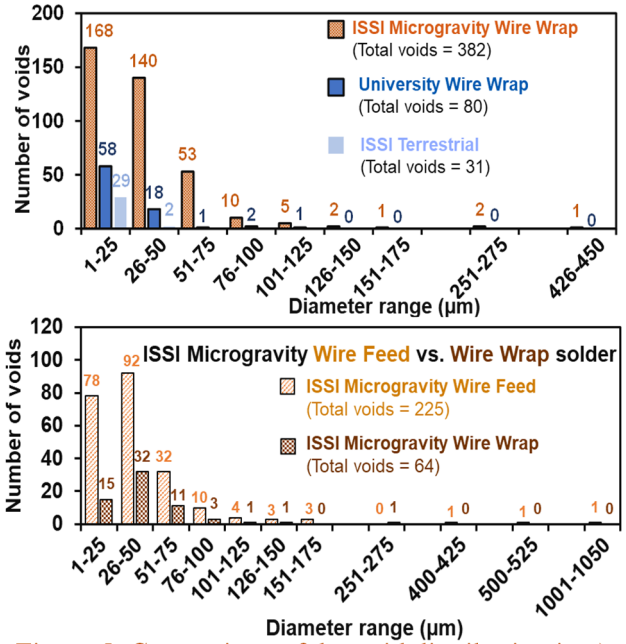


Figure 5: Comparison of the void distribution in a) the ISSI microgravity vs ISSI terrestrial vs. recently prepared terrestrial solder b) the ISSI microgravity wire feed and wire wrap solder. The increased number of voids in the microgravity solder is due to the absence of buoyancy force and

The detailed analysis of voids distribution in the microgravity and terrestrial samples demonstrated that the number of voids in the ISSI microgravity wire wrap solder is ~13 times higher than the ISSI terrestrial solder and ~5 times higher than the recently prepared terrestrial solder (Fig. 5a). The higher void fraction/volume in the ISSI microgravity solder than the terrestrial solder is due to the absence of the buoyancy force on the voids and reduced fluid mixing due to the lack of the Earth's natural convection. Additionally, the voids in the microgravity solder are notably larger than those in the terrestrial solder. The dissimilarities between the ISSI terrestrial solder and the recently prepared terrestrial solder could be due to operator error, as the former was prepared by NASA scientists and the later was prepared by the less experienced operator. In addition, ISSI microgravity wire feed solder demonstrated ~3.5 times higher amount of voids than the ISSI microgravity wire wrap solder (Fig. 5b). During the ISSI experiments, it was observed that the amount of the residual flux on the solder ball was lower in the wire feed samples than the wire wrap samples. This flux behavior resulted from the fact that the rosin flux in the solder core could exit and remain on the solder surface when the solder was free to melt around the Cu wire in the wire wrap solder. In contrast, when the solder wire was fed externally onto a heated Cu wire, and a small amount of liquid solder accrued, the rosin flux from the solder core was constrained to remain internal to the liquid [1]. This observation indicated that the higher flux volume was trapped internally in the wire feed than in the wire wrap solder, which caused a higher void amount.

Microstructure of microgravity vs. terrestrial solder: In terrestrial conditions, heat and mass transfer are affected by gravity-induced buoyancy convection. However, in the *absence* of the Earth's natural convective flow, fluid motion is greatly reduced, driven mainly by solidification volume change, thermally induced density gradients, and Marangoni convection. Thermocapillary or Marangoni convection is a surface tension-driven flow in the presence of the temperature

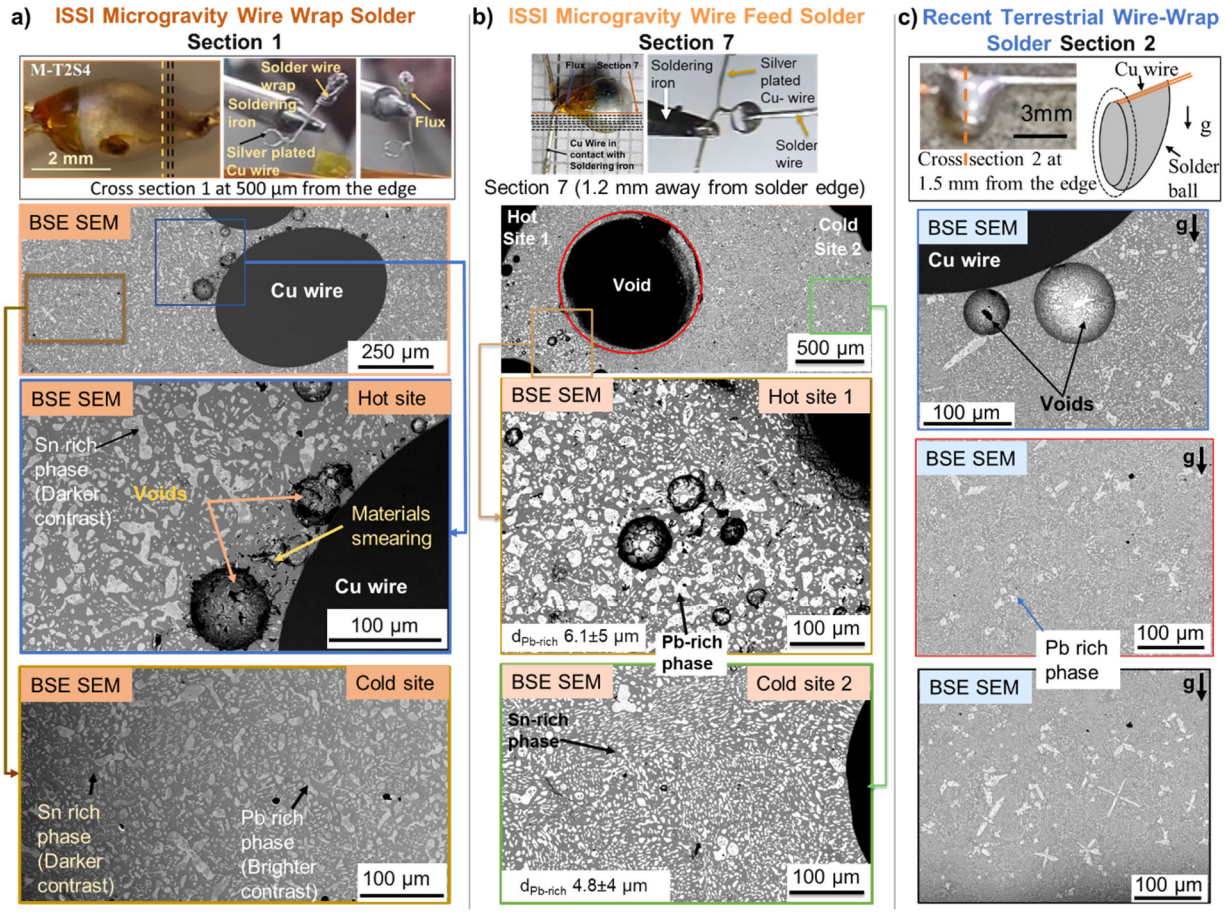


Figure 6: Comparative analysis of a) ISSI microgravity wire-feed sample b) ISSI microgravity wire wrap sample c) Recent terrestrial wire-wrap solder. Microgravity wire feed solder demonstrate increased void and Pb-rich dendrites accumulation than Microgravity wire wrap and terrestrial wire wrap.

gradient [45, 46], causing the liquid to flow from the low surface tension region to the high surface tension region. The effect of this Marangoni convection is less significant in terrestrial conditions due to dominating natural convection. However, the effect of Marangoni flow becomes more significant in microgravity conditions where the bulk natural convection is eliminated. Therefore, to further characterize the effects of the lack of Earth's natural convective flow and buoyancy effects on the microgravity solder microstructure, we performed destructive serial sectioning (Fig. 6a-c). Microstructural characterization demonstrates the presence of the voids close to the Cu wire (Fig. 6a), which may have been at a higher temperature due to the direct contact with the soldering iron.

This higher number of voids near the Cu wire was driven by the presence of the thermocapillary flow. In addition, The ISSI microgravity wire wrap solder demonstrated a homogeneous distribution of Pb-rich phase (Fig. 6a). In contrast to the ISSI microgravity wire wrap solder, ISSI

microgravity wire feed solder demonstrated the accumulation of voids and Pb-rich dendrites in one region (site 1 in Fig. 6b). The average Pb-particle size near site 1 was 6.1 ± 5 µm and near site 2 (expected to be at a higher temperature) was 4.8 ± 5 µm. This difference in the solder microstructure is probably caused by the strong Marangoni flow in the wire feed solder. Further characterization of this sample revealed that this Marangoni effect was probably caused by the presence of a large void (d_{void} = 1.1 mm) in the ISSI microgravity wire feed solder (further discussion in the next section). Additionally, the eutectic lamellae spacing near the hot site was 2.15 ± 0.45 µm, and near the cold site was 1.89 ± 0.43 µm. A slower cooling rate probably caused this lower eutectic lamella spacing at the hot site than at the cold site. In comparison, recently prepared terrestrial solder demonstrated the inhomogeneous distribution of Pb-rich dendrites caused by Earth's gravity, where Pb-rich dendrites (heavier phase) accumulated at the bottom of the solder ball (Fig. 6c).

Marangoni flow due to the large void and thermal convection causes accumulation of smaller voids and Pb-rich dendrites in the higher temperature region in the microgravity solder: To understand what factors contribute to the increased void accumulation (higher number of voids, larger void sizes) microgravity conditions, we considered the effect of the thermal convection and Marangoni flow. The results of the microgravity soldering experiments can be considered as a benchmark for developing solidification theories [47-50]. The improved microstructural homogeneity [51] observed in the microgravity experiments could be used to refine existing theories about the solidification process and to develop new theories that better account for the effects of gravity. However, the effect of Marangoni convection has been overlooked as insignificant under terrestrial conditions due to the dominating natural convection. In the presence of voids, a gradient in the surface tension (σ) at the free surface of the void can induce the Marangoni convection, which can influence the microstructure formation, void mobility [24, 47], dendrite growth [52, 53], heat transfer [45, 54], solid-liquid interface morphology and segregation [55, 56], and particle accumulation [57, 58]. Since the variation of surface tension in solder with temperature is negative, the molten solder at higher temperature has a lower surface tension. Therefore, variation in surface tension at the free surface of the void induces a Marangoni force on the molten solder that causes the liquid flow from the high temperature (low σ) to the low temperature (high σ).

The magnitude of the Marangoni flow due to the void depends on the bubble diameter [59]. Since site 1 (Fig. 6b) in the ISSI microgravity feed solder is expected to be at a higher temperature than site 2, near the soldering iron contact point, there is expected to be a temperature gradient. To calculate the approximate temperature gradient, we calculated the average heat transfer Q using the following equation:

$$Q\ (\mathrm{J}) = mC_p(T_p - T_m) + mh_f \quad \ldots\ldots\ldots (3)$$

The first term in the equation 3 is the heat required to increase the temperature of the molten solder above

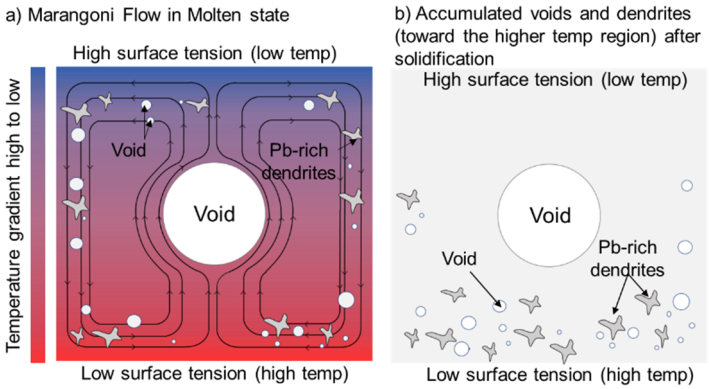


Figure 7: Illustration of a) Marangoni flow due to surface tension gradient around a large void, and b) Pb-rich dendrites and small voids (toward the higher temp region) after solidification. This accumulation is caused by Marangoni flow in molten state.

the melting temperature, and 2nd term is the latent heat of fusion. In the equation above, C_p is the specific heat capacity (0.173 J/g-K) of Pb-Sn solder, T_p = peak temperature based on the ISSI terrestrial soldering experiments (225°C), T_m is the solder melting temperature (183°C), m is the solder mass (0.25 gm) and h_f is the latent heat of fusion (37 J/g). It was assumed that heat is transferred only via conduction from the Cu wire, and heat transfer due to convection and radiation is neglected. Using the equation above, the average heat transfer was calculated to be 11.2 J, and the average heat transfer rate (Q/t) = 1.02 J/s. For a solder ball of 3.8 mm diameter, the temperature gradient can be calculated using the heat conduction equation:

$$Q\ (J)\ =\ -k_{Cu}A_{Cu}\frac{dT}{dx} \ldots\ldots\ldots (4)$$

Using the thermal conductivity of Cu (385 W/m-K) and the diameter of Cu wire (0.41 mm), we calculated the temperature gradient, which has a value of $\frac{dT}{dx}$ = 20010.80 K/m. The temperature difference from the solder ball's hot end to the solder ball's cold end was calculated to be 76 °C. This thermal gradient around the large void (red circle in Fig. 6b, d_{void} = 1.1 mm) causes the surface tension gradient that induces the Marangoni flow, as illustrated in Figure 7. Using the equations below, we calculated the Marangoni number (Ma) and Reynolds number using the following equations:

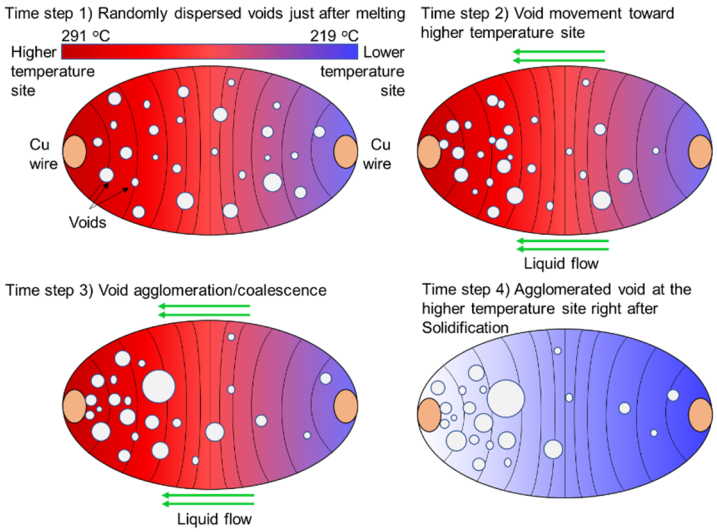


Figure 8: Illustration of the effect of thermal convection on the void and Pb-rich dendrites accumulation. time step 1: void distribution in the microgravity solder right after melting, time step 2: Void movement due to thermal convection induced by thermal gradient, time step 3: coalescence of voids to form large voids, time step 4: distribution of voids right after solidification.

Marangoni number (Ma): $\frac{Marangoni\ transport\ rate}{Diffusive\ transport\ rate} = -\frac{d\sigma}{dT}\left(\frac{L\times dT}{\mu\times\alpha}\right) = 223$

Reynolds number (Re): $\left(\frac{\rho\times u\times L}{\mu}\right) = 1832$

L = characteristic length, u= Flow speed, μ = dynamic viscosity, α = thermal diffusivity, dT = temperature gradient

Both the Marangoni number and Reynolds number are significantly high which demonstrate that the Marangoni flow is significantly high and turbulent. The strong Marangoni flow due to the large void moves the smaller voids toward the higher temperature region (Fig. 7a,b). Furthermore, Marangoni force due to surface tension gradient around the voids will tend to pull them towards the high-temperature region such as the substrate. Additionally, such Marangoni flow causes the Pb-rich dendrites segregation at the high temperature region (Fig. 7b). This causes a shift in the local concentration condition towards higher Pb off-eutectic solder composition, which further results in a higher number of Pb-rich dendrites.

Thus, the Marangoni flow around the large void causes smaller voids and primary Pb-rich dendrites to be self-assembled at the base of the large void. Another reason for the void accumulation is probably thermal convection due to the large thermal gradient within the solder ball. This thermal convection flows the liquid from the cold site to the hot site. Figure 8 illustrates the effect of thermal convection on the void movement. As the solder melts, the flux evaporates and forms the voids which are distributed across the solder cross-section (Fig. 8.1). However, a large thermal gradient in the molten solder induces the thermal convection (Fig. 8.2) that moves the liquid from the cold site to the hot site. This thermal convection moves the voids towards the higher temperature region (Fig. 8.3), where they coalescence and form larger voids (Fig. 8.4).

Task D: Characterizing the effects of the extreme elevated and cryogenic temperature on solder-joint properties:

Long-duration space exploration missions such as Mars rover missions have traditionally kept their electronics in a heated enclosure known as a "warm electronics box" (WEB) to protect them from the extreme cryo temperature on Mars [20]. However, future Mars missions are planning to use a distributed, non-heated architecture for the rover's electronics, where the components will be placed near the systems they interact with rather than in a heated enclosure [20]. The new approach relies on low-cost, commercially available, and space-qualified electronic packaging assemblies. However, these components must be capable of operating in a functional range of – 120ºC to +25ºC for the Mars missions, – 230ºC to +120ºC for the missions to Moon, and – 157ºC to +121ºC outside ISS. Therefore, this project will focus on studying the properties of 1) Pure Tin, 2) 40Pb-60Sn, 3) 50Pb-50Sn, and 4) 95.5Sn-3.8Ag-0.7Cu solder alloys at a) RT, b) Elevated temperatures, and c) Cryo temperatures.

Investigate the effect of aging on microstructure and properties of solders under ambient conditions:

Ex-situ nanoindentation to understand the effect of room temperature aging on near eutectic 40Pb-60Sn solder properties: Due to the high homologous temperature (T_{RT}/T_M = 0.65) for Pb-Sn solder (T_M = 183ºC), classical elemental diffusion described by Fick's law initiates the grain coarsening [60] that can affect the performance and reliability of these solder joints for space applications. Therefore, after over 17 years in storage, the ISSI aged Pb-Sn solder samples are expected to exhibit grain coarsening, softening, and a thick IMC layer. Since aging happens in the solid state of the samples, aging behavior in microgravity solder is expected to be the same as terrestrial solder. Therefore, to account for the effect of aging in 17-year-old microgravity solder, we prepared fresh terrestrial solder samples using a similar method (Wire feed) as the ISSI experiments in order to study the aging properties of Pb-Sn solders at RT.

To determine the effect of aging on 40Pb-60Sn solders properties, two aging conditions were considered: 1) In **surface aging, the** Solder sample was cut, and the same cut surface was examined throughout the aging study, 2) In **bulk aging**, Intact solders were cut on the particular day of the aging and examined. Nanomechanical testing showed a continuous decrease in the solder Berkovich hardness (Fig. 9a) over the aging period, where hardness reduced from 0.19 GPa on day 1 to 0.15 GPa on day 483. This drop in hardness is attributed to the grain coarsening. The

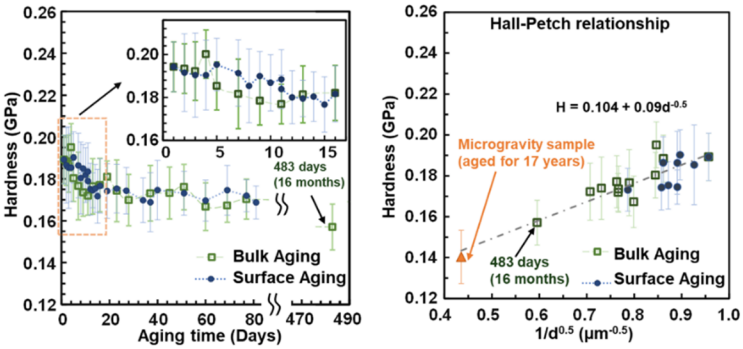


Figure 9: Effect of aging on the hardness of the terrestrial and microgravity solder. The significant drop in hardness and increase in the Pb-particle size in the µg solder is due to longer aging time.

rate at which hardness decreases is higher during the first 15 days of aging after which the age-softening rate goes down. It is also demonstrated that the average hardness drop in the bulk aging condition is faster than the surface aging, which is shown by the higher slope of the bulk aging curve. The microgravity solder (aged for 17 years) demonstrates a similar drop in hardness to recently prepared terrestrial solders as it is on the same hardening slope (Fig. 9b). The significant drop in hardness caused by the larger Pb-particle size in the microgravity solder is believed to be due to the long-term grain coarsening than the recently prepared terrestrial solders. There could be three possible mechanisms that control the grain coarsening, first mechanism is classical diffusion, where the high homologous temperature for the eutectic Pb-Sn provoke the diffusion of Pb and Sn elements which initiates the grain coarsening. The second mechanism is thermo-mechanical stress, where stresses due to the differences in the thermal expansion lattice parameters of Pb-rich and Sn-rich phases also help in grain coarsening. And the third mechanism is surface tension which controls the shape and orientation of the interface in the coarsened grain [9, 60].

Ex-situ nanoindentation to understand the effect of room temperature aging on near eutectic 50Pb-50Sn solder properties:

Future off-eutectic 50Pb-50Sn samples produced aboard the ISS will take many months to return to the Earth for testing. However, off-eutectic solder has a high homologous temperature (T_{RT}/T_M = 0.60) and is expected to experience solid-state diffusion and grain coarsening at room temperature. Therefore, to account for the effect of aging in the microgravity, we prepared fresh terrestrial off-eutectic solder samples to study the rate and rate and effect of aging on this composition.

Three samples were prepared by melting small solder disks into copper coupons in a furnace. Once removed, the samples were allowed to cool to room temperature and were polished to three different faces: top, middle, and vertical (Fig 10). Three different faces were chosen to determine the effects of gravity on aging. Lead is denser than tin, so we expected to see an increase in lead concentration near the bottom of the sample, which likely leads to slightly different aging. Additionally, the effects of the copper substrate were measured by dividing the top and middle samples into two regions: the center and the edge.

Testing was done using nanomechanical indentation tests over the course of 90 days in addition to SEM imaging to record the grain size evolution. Every testing day involved performing 36 indentation tests in each region to measure local surface hardness. Additionally, the samples were imaged using an SEM to record where the previous day's indents landed as well as take images for grain size measurements. In each region of the samples, at least three images were taken for grain size analysis. Images were also taken at control points on each sample to visually inspect grain coarsening throughout the experiment. Testing was completed every day for the first 15 days after the samples were solidified, then every three days from 15 days to 60 days, then every six days from 60 days to 90 days.

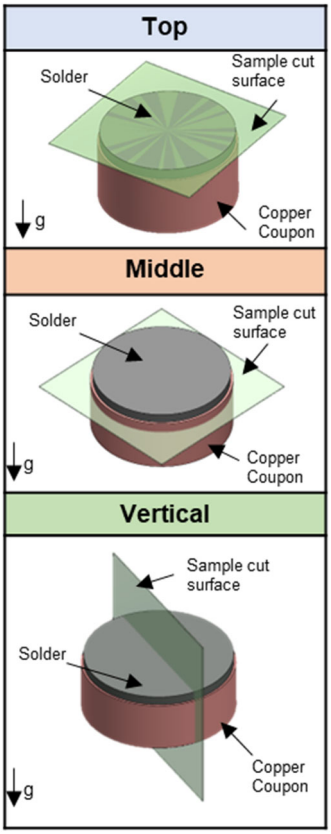
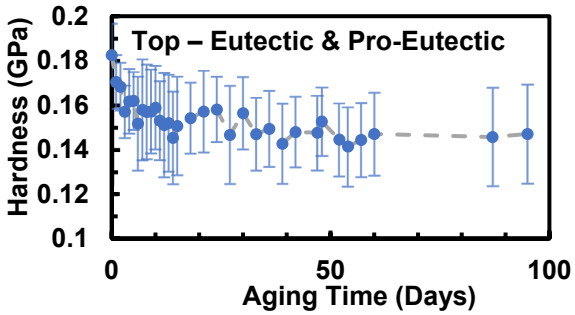


Figure 10: Depiction of the cut planes of each of the three samples used for off-eutectic aging.

Off-eutectic solder experiences rapid aging and grain growth at room temperature. Nanomechanical testing showed a decrease in hardness of the top sample from 18.3GPa on the first day to 14.7GPa by day 90 (Fig 11). The off-eutectic hardness was found to be lower than the eutectic hardness due to the increased amounts of lead. Within the off-eutectic samples, the Pb-Rich regions showed reduced hardness compared to the eutectic regions. Additionally, the hardness decreased the fastest over the first 15 days after solidification, similar to the eutectic aging rate.



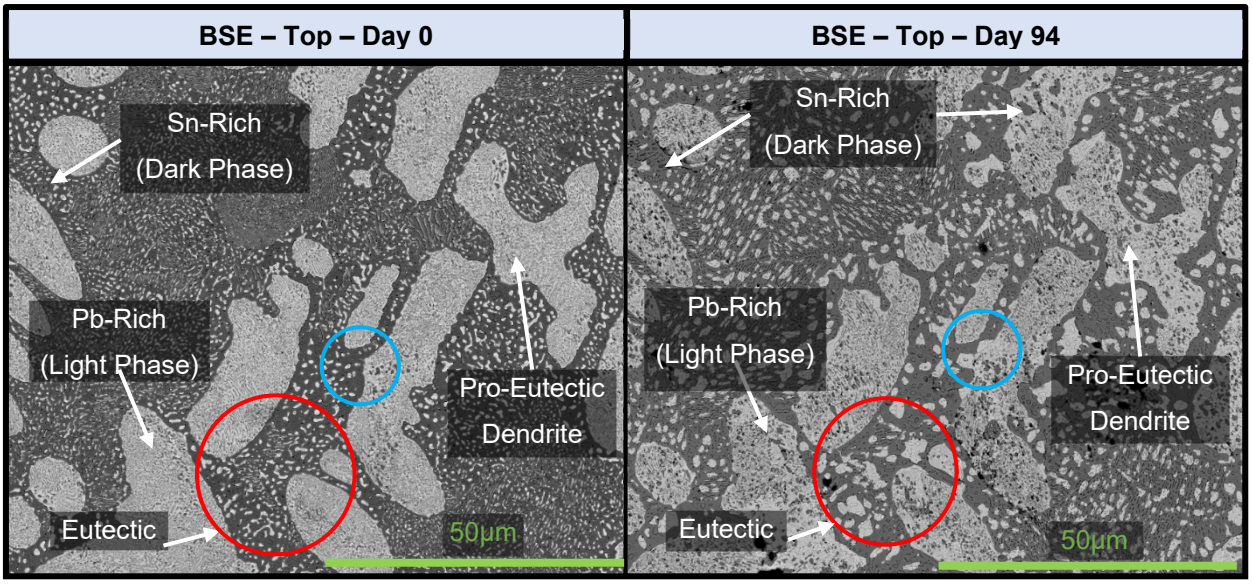


Figure 12: Comparison of the top section control region between day 0 of the experiment and day 94. Clear grain growth is occurring in the eutectic phase as well as Sn-Rich precipitation within the Pb-Rich pro-eutectic dendrites. The red circles outline grain growth in a part of the eutectic region and the blue circles outline grain growth along the edge of an Pb-Rich dendrite.

Grain coarsening is visible in both eutectic and pro-eutectic regions, as well as both the Pb-Rich and Sn-Rich phases (Fig 12). In addition to grain coarsening, Sn-Rich precipitates form within Pb-Rich regions due to significantly lower solubility of Sn in Pb at room temperatures than at the eutectic temperature. The rate of grain coarsening was measured by finding the average area of Pb-Rich phases within the eutectic structure and calculating the diameter of a circle with the same area (Fig 13). Similarly, to hardness, the grains grew rapidly over the first 15 days, then eventually the average grain size leveled out. Additionally, the error bars start with a range of 0.6µm at day 0, but they grow to 1.1µm by day 90. It is expected that the error bars grew quite rapidly due to some of these grains shrinking as they give up volume, and others are growing as they gain

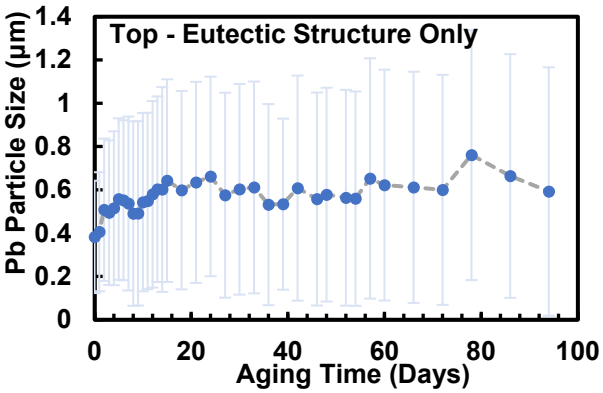


Figure 13: Average equivalent diameter of Pb-Rich grains in the eutectic region of the top sample over 90 days.

volume. Further analysis of distribution of grain sizes is required to understand the distribution of grain growth.

Generally, structured materials show an inverse relationship between grain size and hardness. Work is being done to show that the off-eutectic composition follows this behavior. Currently, indentation data is not separated based on the region it lands in, and grain size measurements are only taken from the eutectic region. However, plotting the eutectic grain size against the eutectic and pro-eutectic hardness shows a definite scaling behavior (Fig 14).

Early analysis shows that the middle and vertical sections show similar aging rates to the top section. The middle section shows lower hardness than the top section, and the vertical section shows lower hardness than both other sections due to heavier lead accumulating near the bottom. Ongoing analysis is investigating the effects of gravity and the substrate on the effect of aging.

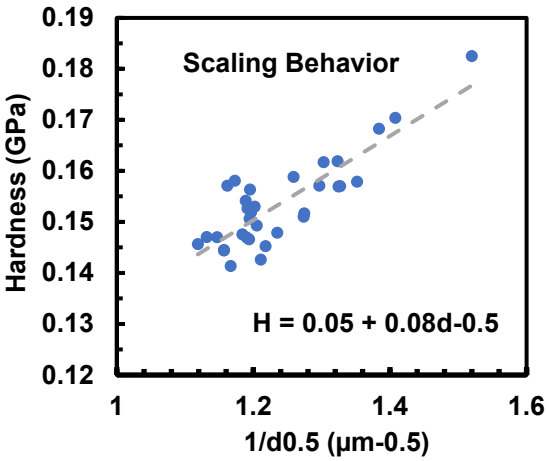


Figure 14: Scaling behavior between the grain size of the off-eutectic solder and the hardness. Grain size data is the average equivalent diameter of Pb-Rich grains in the eutectic region of the top sample. Hardness data is the average hardness of the eutectic and pro-eutectic phase of the top sample.

Investigating the Effect of the cryogenic temperature on β-Sn to α-Sn phase transformation and resultant properties of the solder joints: Sn-based solders undergo a ductile to brittle transition (DBT) at the cryo temperature leading to brittle crack formation, which can cause the catastrophic failure of the solder joints. Additionally, Sn-based solders generally occur a β-Sn to α-Sn phase transition below 13°C, where β-structure (Body-centered tetragonal) transforms to α-structure (Diamond cubic).

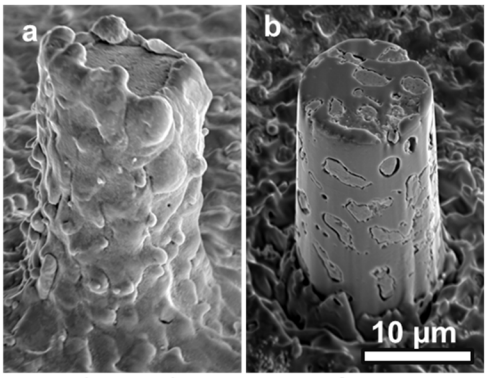


Figure 15: a) rough-cut 40Pb-60Sn micropillar using the micro-EDM b) final pillar after the pillar cleaning using the focused ion beam milling at very low current (0.85 nA).

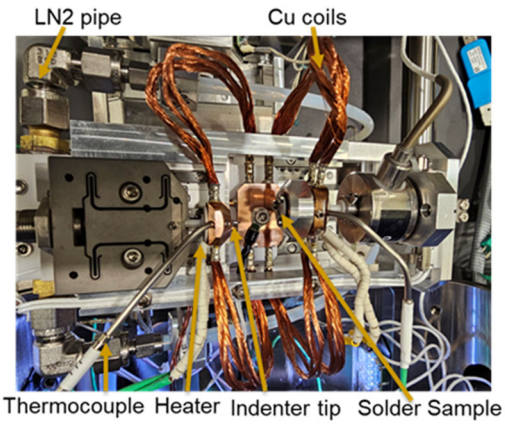


Figure 16: Alemnis in-Situ SEM nanoindenter used for micro-pillar compression experiments at cryogenic temperatures.

This phase transformation causes a 26% increase in volume that generates internal stresses, due to which the reliability of the Sn-based solder joints is compromised for space applications. Therefore, to investigate the effect of various cryo temperatures (similar to temp conditions on the Moon, Mars, and outside ISS) on

a β-Sn to α-Sn phase transition, we performed in-situ SEM micropillar compression on 40Pb-60Sn solder at RT, – 40ºC, – 85ºC, – 120ºC and – 145ºC. These temperatures are selected based on the literature review [21, 22, 61] on Sn phase transformation and solder joint application during space exploration missions.

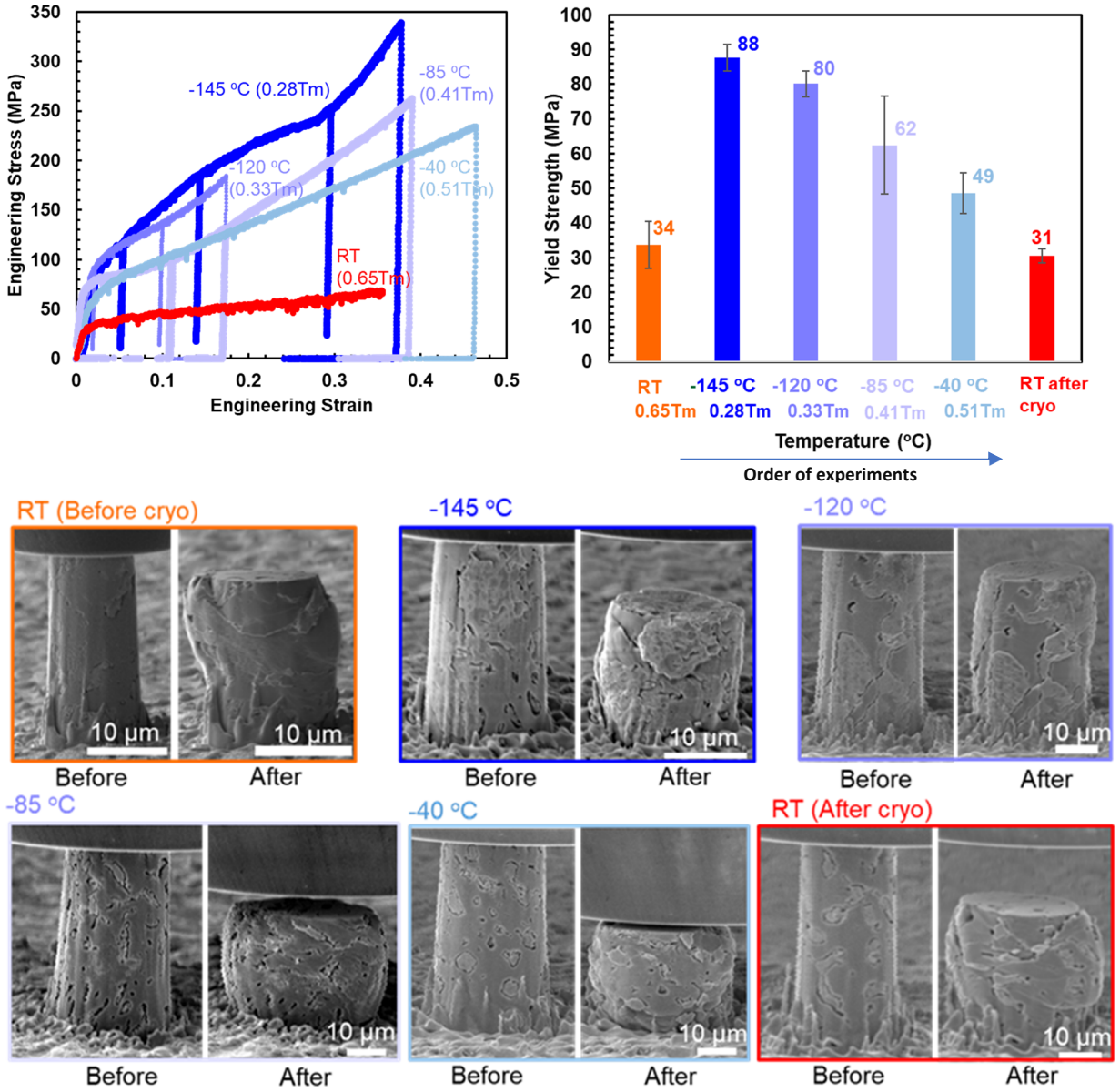


Figure 17: a) Engineering stress-strain from the 40Pb-60Sn solder at -145 ºC, -120 ºC, -85 ºC, -40 ºC and RT b) 0.2% offset yield strength c) SEM images of micropillars before and after the compression. The strength of micropillars is higher at cryo temperatures than the room temperature.

We fabricated the rough cut of the micropillars (Fig. 15a) on 40Pb-60Sn using the micro-EDM at the Small-Tec. The rough micropillars were cleaned (Fig. 15b) using the focused ion beam milling at a very low current (0.85 nA). The micropillars were then compressed at the loading rate of 75nm per seconds at room and various cryogenic temperatures using the Alemnis in-Situ SEM cryo indenter (Fig. 16). During the compression, load, displacement and time were recorded with simultaneous SEM imaging and video recording. The load displacement were converted into the Stress-strain and the 0.2% offset yield strength was calculated.

As the temperature reduces, solder demonstrate higher hardening (Fig. 17a) and strength (Fig. 17b) under the compressive loading conditions. This increase in the solder strength at the cryo temperatures is probably caused by the lack of dislocation mobility due to low thermal activation energy. Figure 17c demonstrate the SEM images of pillars at various cryogenic temperatures before and after compression experiments. The in-depth data analysis of these experiments is still under investigation to understand the effect of ductile to brittle transition and beta Sn to alpha Sn transformation on the solder properties.

Future work:

Investigating the effect of accelerated aging on microstructure and properties of solders under elevated temperatures typically experienced by solder joints in space:

We will investigate the long-term reliability of the solder joints under various elevated temperature conditions. We will utilize the high-temperature capabilities of the Alemnis In-Situ SEM nanoindenter to obtain the nanoindentation stress-strain from different phases (e.g., Pb-rich and Sn-rich) at temperatures ranging from RT to 120°C. The testing temperature and corresponding homologous temperatures for different solder compositions are given in figure 19b. Our novel high-throughput spherical nanoindentation stress-strain data-analysis protocols [62-64] which allow us to measure both the elastic (e.g. modulus) and plastic (indentation yield Y_{ind} (Fig. 18a) and strain hardening coefficients) properties from indentation zones ranging in size from ~30 nm to 10 µm [62, 63]. By using an indentation zone size smaller than the critical feature size [65], we can measure changes in Y_{ind} as a function of orientation [63, 66]. Similarly, by using a sufficiently large sized indenter we propose to interrogate the bulk properties (eg. combined response of Pb-rich and Sn-rich phases) of the solder as a function of temperature.

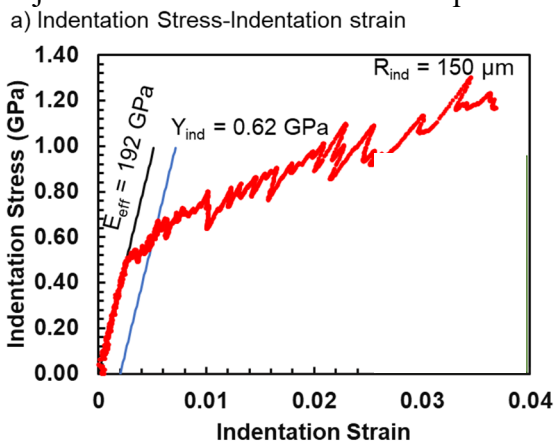


Figure 18: Spherical nanoindentaion on Ta(111) to obtain initial elastic, elastic to plastic and post yield behavior.

We expect higher grain coarsening and higher intermetallic growth caused by a higher elemental diffusion rate at the high temperature, which is expected to increase with increasing temperature. Furthermore, we expect the lower Y_{ind} at higher temperatures caused by the larger grain size as coarsened phases cannot effectively block the dislocation movement.

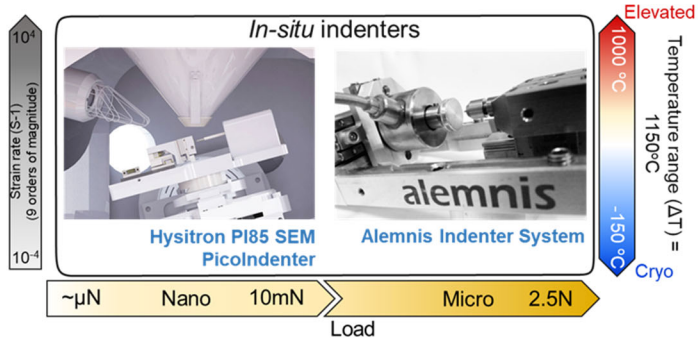
In-situ SEM micromechanical testing under extreme cryo temperature conditions:

We will perform high throughput spherical indentation to obtain yield strength and post-yield hardening behavior from three different solder compositions a) Near eutectic 40Pb-60Sn b) 50Pb-50Sn, and c) 95.5Sn-3.8Ag-0.7Cu at cryogenic temperatures mentioned in figure 19b. In addition to the high throughput spherical nanoindention, we will utilize specialized techniques such as micro-pillar compression (Fig. 19c) and micro-tensile experiments (Fig. 19d) to characterize the compressive and tensile strength and strain at failure. These tests can be repeated at various length scales, in order to interrogate the local vs. bulk behavior of the material. However, these experiments are costly and time-consuming. It also requires minimal taper (<10%) and perfect indenter probe and sample alignment to get reliable data. Therefore, high throughput spherical nanoindentation is our primary choice to get Y_{ind} and post-yield hardening behavior, followed by

micropillar compression and mico-tensile experiments to understand the change in the failure mechanism during DBT at cryo temperatures. We expect lower strain to failure and lower toughness in lead-free 95.5Sn-3.8Ag-0.7Cu solder than the Pb-Sn solder. We also expect higher DBTT for lead-free 95.5Sn-3.8Ag-0.7Cu solder than Pb-Sn solder. In 50Pb-50Sn solders, we expect higher toughness and higher strain to failure, and lower DBTT due to higher content of Pb than the 40Pb-60Sn.

Microstructural characterization of microgravity ISSI solders:

We will perform destructive serial sectioning on remaining microgravity solder, particularly focusing on Microgravity ISSI test 4 sample 16, test 2 sample 10 and test 2 sample 16.



Testing temperature (°C)	Homologues Temperature (T_{test}/T_m)					
	40Pb-60Sn/ 50Pb-50Sn	95.5Sn-3.8Ag-0.7Cu	Sn	Pb	Ag	Cu
25	0.65	0.61	0.59	0.50	0.24	0.22
50	0.71	0.66	0.64	0.54	0.26	0.24
75	0.76	0.71	0.69	0.58	0.28	0.26
100	0.82	0.76	0.74	0.62	0.30	0.27
125	0.87	0.81	0.79	0.66	0.32	0.29
0	0.60	0.56	0.54	0.45	0.22	0.20
-40	0.51	0.47	0.46	0.39	0.19	0.17
-85	0.41	0.38	0.37	0.31	0.15	0.14
-145	0.28	0.26	0.25	0.21	0.10	0.09

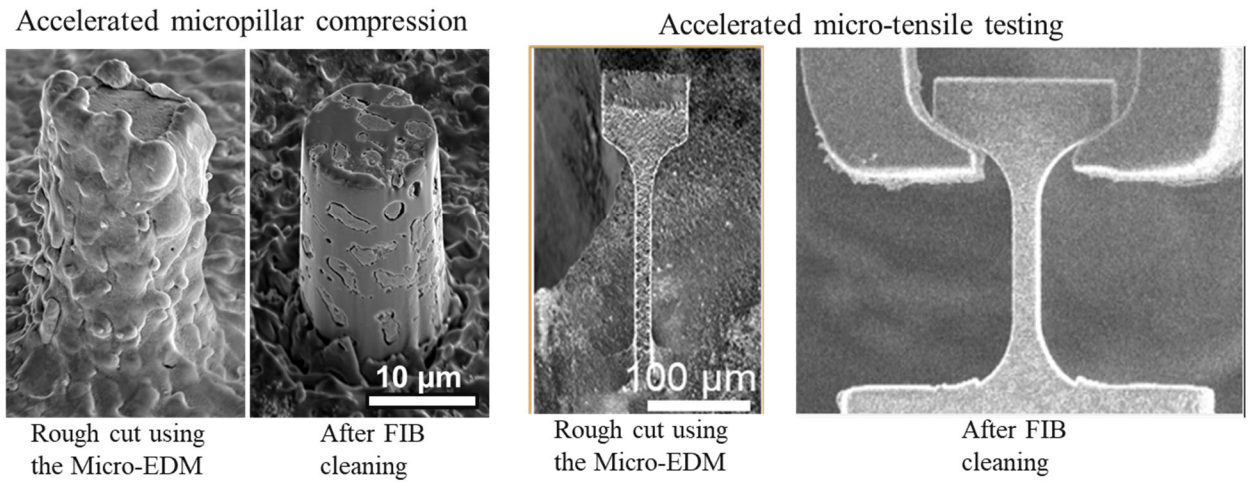


Figure 19: a) Cryo and elevated temperature Micro-mechanical testing facilities in our lab, b) The testing temperature and corresponding homologous temperatures for different solder compositions, c) Accelerated micropillar compression experiments, d) Accelerated micro-tensile experiments.

Awards

Manish Kumar, a graduate student working on this NASA PSI project was awarded with the following awards:

- ASGSR Student Travel Award ($500) - to help defray the costs of attendance at the American Society for Gravitational and Space Research (ASGSR) conference 2023.
- Wayne G. Basler Scholarship ($5000) for 2023 in Materials Science and Engineering at Iowa State University.

- Sigma Xi Grants ($1000) in Aid of Research 2023 - Among 8% of the total applicants (~1000).

Following UG students working on this project received various awards:

- Joshua Kamp and Noah Madison were awarded First-Year Honors Mentor Program Grant-2023.
- Noah Madison was awarded with the Outstanding First-Year Honors Mentor Program (FHMP) Scholar Award – 2023.
- Undergraduate research assistant Soren Hellyer has been awarded the TMS Acta Materialia scholarship for the 2024 calendar year.

Conferences:

The Minerals, Metals, and Material Society 2023 Conference (TMS-2023)

Manish Kumar gave an oral presentation at The Minerals, Metals, and Material Society 2023 Conference (TMS-2023), San Diego, California. His presentation was titled "Structure and Properties of Solder Joints Produced in Terrestrial and Microgravity Conditions."

International Space Station Research and Development conference 2023:

Manish Kumar gave a poster presentation on "Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures" at the International Space Station Research and Development conference (ISSRDC – 2023). The event was hosted by the Center for the Advancement of Science in Space (CASIS), manager of the International Space Station (ISS) U.S. National Laboratory; NASA; and the American Astronautical Society.

American Society of Gravitational and Space Research Conference 2023

Manish Kumar gave an oral presentation on "Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures" at the American Society for Gravitational and Space Research conference (ASGSR-2023) in Washington, DC. Manish was also awarded the ASGSR Student Travel Award-2023 to help defray the costs of his attendance at the ASGSR-2023 conference.

Center for Advanced Non-Ferrous Structural Alloys (CANFSA) IRB Review meeting 2023

Dr. Pathak presented a poster on "Structure and Properties of Pb-Sn Solders Produced in Terrestrial vs. Microgravity Environments" at the Center for Advanced Non-Ferrous Structural Alloys (CANFSA-2023) IAB Review Meeting, Colorado School of Mines.

National Conference on Undergraduate Research (NCUR-2023)

UG students Soren Hellyer and Caleb M Ogden presented on "Studying the Effects of Aging on the Structure and Properties of Off-Eutectic 50wt%Pb-50wt%Sn Solder Joints for In-Space Applications" at the National Conference on Undergraduate Research (NCUR-2023) in Eau Claire, Wisconsin.

The Symposium on Undergraduate Research & Creative Expression-2023

UG students Joshua Kamp and Noah Madison presented on "Structure and Properties of Pb-Sn and Pb-Free Sn-Ag-Cu Solders Produced in Terrestrial Environment" at the Symposium on Undergraduate Research & Creative Expression-2023 organized by Iowa State University, Ames.

Conference presentation slides:

Please see the attached slides:

Attachment 1_ASGSR_Manish Kumar NASA Soldering_2023-11-16
Attachment 2_Hellyer_Ogden_ASGSR_SolderAging_Final

Monthly reports:

NASA PSI (**80NSSC23K0279**) January 2023 report

Topic 1. We performed microstructural analysis and nanomechanical testing on the ISSI microgravity (Test 4 sample 16, small loop wire feed, cross-sections 3 and 4), ISSI terrestrial (small loop wire feed, Cross-section 3 and 4). We calculated void content, void distance from Cu wire, and Pb-particle size. We also prepared a terrestrial solder (small loop wire feed) sample in our lab for a comparative study with ISSI solders.

Topic 2: Aging study on Off eutectic Pb-Sn (50wt%Pb-50wt%Sn) solder samples is undergoing and has been completed for the first 60 days. We measured hardness and Pb-particle size as a function of the aging period.

We have also received two following awards:

1. Undergraduate students Soren Hellyer and Caleb M Ogden, working with Dr. Pathak, have received funding from NASA – Iowa Space Grant Consortium (ISGC) to study the Effects of Aging on the Structure and Properties of Off-Eutectic Pb-Sn Solder Joints for In-Space Applications. The goal of the NASA – Iowa Space Grant Consortium (ISGC) is to provide experiences to undergraduates that promote skills for STEM careers through hands-on activities.
2. We have received the Iowa NASA EPSCoR Partnership Development Travel grant. This grant is to support the travel to NASA International Space Station (ISS) implementation partners - Tec-Masters Inc (TMI), Huntsville, AL, to perform metal joining experiments and Advanced Photon Source (APS), Lemont, IL, for conducting high-resolution X-ray tomography experiments for 3-D microstructural characterization.

Experimental plan for February 2023:

Topic 1. We have been allotted beamline time at the Advanced Photon Source on February 8, 2023. We will perform non-destructive high-resolution and high-speed tomography experiments on the solder samples to get a 3-D distribution of grains and porosity.

Topic 2. Microstructural characterization and nanomechanical testing using destructive serial sectioning on the ISSI microgravity, ISSI terrestrial, and ISU terrestrial solder samples (wire feed solder samples).

NASA PSI (**80NSSC23K0279**) February 2023 report

Topic 1. We performed non-destructive high-resolution and high-speed tomography experiments on the solder samples at the Advanced Photon Source (APS) to get a 3-D distribution of grains and porosity.

Topic 2. We performed microstructural analysis and nanomechanical testing on the ISSI microgravity (Test 4 sample 16, small loop wire feed, cross-sections 5), ISSI terrestrial (small loop

wire feed, Cross-section 5). We calculated void content, void distance from Cu wire, and Pb-particle size.

Topic 3: Aging study on Off eutectic Pb-Sn (50wt%Pb-50wt%Sn) solder samples is undergoing and has been completed for the first 85 days. We measured hardness and Pb-particle size as a function of the aging period.

Plan for March 2023:

Topic 1. Training undergraduate students for 3-D data reconstruction from the raw data obtained at APS and data analysis using Dragonfly software.

Topic 2. Microstructural characterization and nanomechanical testing using destructive serial sectioning (Section 6) on the ISSI microgravity, ISSI terrestrial, and ISU terrestrial solder samples (wire feed solder samples).

Topic 3. We will present our work on this project at the Materials Research in Reduced Gravity symposium at the TMS-2023, San Diego conference on March 22nd. The presentation schedule is as follows:

Meeting: 2023 TMS Annual Meeting & Exhibition
Symposium: Materials Research in Reduced Gravity
Session: Solidification
Session Chair(s):
Date: 03/22/2023
Time: 4:20 PM - 4:40 PM
Location: San Diego Convention Center - 30B
Designated speaker: Manish Kumar

NASA PSI (**80NSSC23K0279**) March 2023 report

Topic 1. We presented our work on this project titled "Structure and Properties of Pb-Sn solders produced in Microgravity and Terrestrial Environments" at the Materials Research in Reduced Gravity symposium at the TMS-2023, San Diego conference on March 22nd.

ppt: https://iastate.box.com/s/8zs60uq3dlko1k3rpqy5y1nn0mtmuud6

pdf: https://iastate.box.com/s/eq0ey10paqy4fkya5tkwjh1iq9040cll

Topic 2. We performed microstructural analysis and nanomechanical testing on the ISSI microgravity (Test 4 sample 16, small loop wire feed, cross-sections 6). We calculated void content, void distance from Cu wire, and Pb-particle size.

Topic 3: Aging study on Off eutectic Pb-Sn (50wt%Pb-50wt%Sn) solder samples is undergoing and has been completed for the first 115 days. We measured hardness and Pb-particle size as a function of the aging period.

Topic 4: We worked on uploading the NASA PSI soldering project data to the NASA PSI box folder.

Plan for April 2023:

Topic 1. Microstructural characterization and nanomechanical testing using destructive serial sectioning (Section 7) on the ISSI microgravity, ISSI terrestrial, and ISU terrestrial solder samples (wire feed solder samples Section 5 and 6).

Topic 2. Presentation on "Effect of Aging on Structure and Properties of Off-Eutectic 50Pb-50Sn solder for In-Space Applications" by UG students Soren Hellyer and Caleb Ogden at The National Conference on Undergraduate Research (NCUR), April 13-15, 2023.

NASA PSI (**80NSSC23K0279**) April 2023 report

Topic 1. Undergraduate students Soren Hellyer and Caleb Ogden presented their work on "Effect of Aging on Structure and Properties of Off-Eutectic 50Pb-50Sn solder for In-Space Applications" at The National Conference on Undergraduate Research (NCUR), April 13-15, 2023.

Link: https://iastate.box.com/s/400nmg6eote3ne4b4akyn8k4tww7d3qe

Topic 2. We performed microstructural analysis and nanomechanical testing on the ISSI microgravity (Test 4 sample 16, small loop wire feed, cross-sections 7), ISSI terrestrial (sample 2, section 4 to 6), ISSI microgravity (Test 2 sample 5, wire wrap solder, section 1 and 2). We calculated void content, void distance from Cu wire, and Pb-particle size.

Topic 3. Undergraduate student Noah Madison and Joshua Kamp presented their work on "Structure and Properties of Pb-Sn and Pb-Free Sn-Ag-Cu Solders Produced in Terrestrial Environment" at the Symposium on Undergraduate Research & Creative Expression-2023 organized by Iowa State University, Ames.

Link: https://iastate.box.com/s/x83n88ggsebq4hyy3x59eispmrxzvtwu

Plan for May 2023:

Topic 1. Microstructural characterization and nanomechanical testing using destructive serial sectioning (Section 8) on the ISSI microgravity, ISSI terrestrial, ISU terrestrial solder samples (wire feed solder samples Section 7), and ISSI microgravity (Test 2 sample 5, wire wrap solder, section 3).

Awards: Noah Madison working on this project, was awarded the Outstanding First-Year Honors Mentor Program (FHMP) Scholar Award - 2023

NASA PSI (**80NSSC23K0279**) June 2023 report

Topic 1. Mr. Manish Kumar completed his Ph.D. preliminary oral examination on June 9, 2023. Please see the link below for the presentation (pdf attached to this mail).

ppt: https://iastate.box.com/s/86fplefq6p5ye30us9up75qqn78ejnk7

Topic 2. We performed microstructural analysis and nanomechanical testing on the ISSI microgravity (Test 2 sample 5, wire wrap solder, sections 3 to 8). Data analysis on these sections will be performed in July 2023.

Topic 3. We performed calculations on the Marangoni flow due to the void in the microgravity conditions.

Plan for July 2023:

Topic 1. Presentation on **"Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures"** at the Physical Metallurgy Gordon Research Conference (July 9 to July 14, 2023).

Poster presentation schedule: From 4 pm to 6pm on July 12 (Wednesday) and July 13 (Thursday)

Topic 2: Microstructural characterization and nanomechanical testing using destructive serial sectioning on the ISSI microgravity (Test 2 sample 5, wire wrap solder, sections 9 to 14).

Topic 3. Data analysis (void content, void distance from Cu wire, and Pb-particle size) on the Test 2 sample 5, wire wrap solder, sections 3 to 8.

Topic 4. We plan to prepare the micro-tensile and micro-pillar specimens on the solder samples at the Smal-Tec. These specimens will allow us to investigate the effect of cryo temperature on solder properties.

NASA PSI (**80NSSC23K0279**) July 2023 report

Topic 1. Manish Kumar presented a poster on **"Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures"** at the International Space Station Research and Development Conference (ISSRDC 2023)

Poster: https://iastate.box.com/s/zehgpq0hmkqyxjcjsjfh7hzuawi2mxwk

Topic 2. Manish Kumar presented a poster on **"Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures"** at the Physical Metallurgy Gordon Research Conference (July 9 to July 14, 2023).

Topic 3. We performed microstructural analysis on the 1. ISSI microgravity Test 4 sample 16, wire wrap solder, sections 8 and 9, 2. ISSI microgravity Test 2 Sample 10 Section 3 and 4, 3. ISSI terrestrial Test 2 Section 7 and 8.

Topic 4. UG student Caleb Ogden performed microstructural data analysis on ISSI microgravity Test 2 Sample 5. He calculated void size, void distance from Cu wire, and Pb-particle size.

Plan for August 2023:

Topic 1: Microstructural characterization and nanomechanical testing using destructive serial sectioning on 1. ISSI microgravity Test 4 sample 16, wire wrap solder, sections 10 and 11, 2. ISSI microgravity Test 2 Sample 10 Section 5 and 6, 3. ISSI terrestrial Test 2 Section 9 and 10.

Topic 2. Remaining Data analysis (void content, void distance from Cu wire, and Pb-particle size) on the Test 2 sample 5, wire wrap solder, sections 3 to 8.

Topic 3. We plan to prepare the micro-tensile and micro-pillar specimens on the solder samples at the Smal-Tec in August. These specimens will allow us to investigate the effect of cryo temperature on solder properties.

NASA PSI (**80NSSC23K0279**) September 2023 report

Topic 1. We fabricated 13 mico-pillar specimens on a 40Pb-60Sn solder sample at small-tec, Chicago, IL. We also performed in-situ SEM micro-pillar compression experiments at room temperature using our in-situ SEM Alemnis nanoindenter and analyzed the data.

Topic 2: We polished Sn (100), Sn (110), and Sn(111) single crystals and performed preliminary spherical indentation experiments using a 150-micron indentation radius to obtain indentation stress-strain. These single-crystal experiments will create a baseline for the measurement of our bulk property (combined response of multiple phases).

Topic 3. UG students Caleb Ogden and Soren Hellyer performed aging data analysis on 50Pb-50Sn solder.

Plan for October 2023:

Topic 1: Microstructural characterization and nanomechanical testing using destructive serial sectioning on 1. ISSI microgravity Test 4 sample 16, wire wrap solder, sections 10 and 11, 2. ISSI microgravity Test 2 Sample 10 Section 5 and 6, 3. ISSI terrestrial Test 2 Section 9 and 10.

Topic 2. We plan to prepare the 50 micro-pillar specimens on the 40Pb-60Sn solder samples at the Smal-Tec in October. These specimens will allow us to investigate the effect of cryo temperatures (-150 degree C, -80 dgree C, -45 degree C, -20 degree C) on solder properties.

Awards:

1. Graduate student Manish Kumar, working on the NASA soldering project, was awarded the Wayne G. Basler Scholarship ($5000) for 2023 in the Department of Materials Science and Engineering at Iowa State University. This honor is awarded annually to an MSE graduate student who exemplifies an exceptionally high level of dedication to education, demonstrating "the desire and work ethic it takes to be the best they can be."
2. Undergraduate research assistant Soren Hellyer, working on NASA solder aging project, has been awarded the TMS Acta Materialia scholarship for the 2024 calendar year. He will be awarded the scholarship amount of $5000 at the TMS 2024 Annual Meeting and Exhibition in Orlando, FL. Soren will also receive a $500 travel stipend as a part of the award.

NASA PSI (80NSSC23K0279) October 2023 report

Topic 1: We fabricated 100 micro-pillar specimens on two different 40Pb-60Sn solder samples at small-tec, Chicago, IL. Following the rough cut at the small-tec, we cleaned 40 pillars using the focused ion beam (FIB) milling for our room temperature and cryo temperature experiments.

Topic 2: We performed in-situ SEM micro-pillar compression experiments on 40Pb-60Sn solder at different temperatures on 33 micropillars using our in-situ SEM Alemnis nanoindenter. The targeted temperatures and order of the experiments are as below:

1) Room Temperatures
2) -145 degrees C
3) -120 degrees C
4) -85 degrees C
5) -40 degrees C
6) Room temperature

Topic 3: UG students Noah Madison and Joshua Kamp were trained on Spherical Indentation Stress-Strain techniques. They will utilize this technique to obtain indentation stress-strain from Lead and Tin single crystals. These single-crystal experiments will create a baseline for the measurement of our bulk property (combined response of multiple phases).

Topic 4: UG students Caleb Ogden and Soren Hellyer performed aging data analysis on 50Pb-50Sn solder.

Plan for October 2023:

Topic 1: Oral presentation on "Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures" at the ASGSR 2023 conference in Washington, D.C. Presentation details are as follows:

Presentation Title: Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures

Author: Manish Kumar, Ralph Napolitano, Sid Pathak
Session Name: Materials
Session Day: Saturday, November 18, 2023
Session Time: 1:30 PM - 3:30 PM
Session Room: TBD

Topic 2: Poster presentation on " Studying the Effects of Aging on the Structure and Properties of Off-Eutectic Pb-Sn Solder Joints for In-Space Applications" at the ASGSR 2023 conference in Washington, D.C. Presentation details are as follows:

Presentation Title: Studying the Effects of Aging on the Structure and Properties of Off-Eutectic Pb-Sn Solder Joints for In-Space Applications

Author: Soren Hellyer, Caleb Ogden, Manish Kumar, Sid Pathak
Session Name: Undergraduate and High School Poster Session
Session Day: Saturday, November 18, 2023
Session Time: 10:00 AM - 12:00 PM
Session Room:TBD

Topic 3: 40Pb-60Sn micropillar compression data analysis.

Topic 4: Microstructural characterization and nanomechanical testing using destructive serial sectioning on 1. ISSI microgravity Test 4 sample 16, wire wrap solder, sections 10 and 11, 2. ISSI microgravity Test 2 Sample 10 Section 5 and 6, 3. ISSI terrestrial Test 2 Section 9 and 10.

Awards:

1.Graduate student Manish Kumar, working on the NASA soldering project, was awarded the ASGSR Travel Grant to support his travel to the ASGSR-2023 conference in Washington, D.C.

NASA PSI (**80NSSC23K0279**) November 2023 report

Topic 1: Manish Kumar gave an oral presentation on "Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures" at the ASGSR 2023 conference in Washington, D.C

Topic 2: UG students Soren Hellyer and Caleb Ogden presented a poster on " Studying the Effects of Aging on the Structure and Properties of Off-Eutectic Pb-Sn Solder Joints for In-Space Applications" at the ASGSR 2023 conference in Washington, D.C.

Topic 3: We were invited to the NASA PSI proposers' conference to briefly speak about our experience successfully using existing PSI data to write a proposal.

Topic 4: 40Pb-60Sn micropillar compression data analysis – Calculation of stress, strain, and strain hardening at various cryogenic testing temperatures:

1) Room Temperatures

2) -145 degrees C

3) -120 degrees C

4) -85 degrees C

5) -40 degrees C

6) Room temperature

Topic 5: Fabrication of 40Pb-60Sn micropillar using the Micro-EDM at Small-Tech for high-temperature compression experiments. Also, we cleaned these pillars using the focused ion beam (FIB).

Plan for December 2023:

Topic 1: Polishing of Pb and Sn single crystals and spherical indentation experiments for Indentation Stress-Strain calculations. These experiments will generate a baseline for our future indentation experiments on solder at elevated and cryo temperatures.

Topic 2: In-depth analysis (correlation of Stress-strain and hardening response with SEM videos and images) of 40Pb-60Sn micropillar compression data (obtained at cryo temperatures) to understand the effect of ductile to brittle transition on solder properties.

Topic 3: Fabrication of micro tensile specimens on 50Pb-50Sn solder using the Micro-EDM at Small-Tec for room temperature tension experiments.

Topic 4: Fabrication of micro-pillars on 50Pb-50Sn and 95.5Sn-3.8Ag-0.7Cu solders for cryogenic experiments.

References:

[1] R. Grugel *et al.*, "Final Research Report. In-Space Soldering Investigation (ISSI)," 2006.

[2] R. N. Grugel *et al.*, "Materials research conducted aboard the International Space Station: Facilities overview, operational procedures, and experimental outcomes," *Acta Astronautica,* vol. 62, no. 8, pp. 491-498, 2008/04/01/ 2008, doi: https://doi.org/10.1016/j.actaastro.2008.01.013.

[3] P. J. Wang, J. S. Kim, and C. C. Lee, "Fluxless Bonding of Large Silicon Chips to Ceramic Packages Using Electroplated Eutectic Au/Sn/Au Structures," *Journal of Electronic Materials,* vol. 38, no. 10, pp. 2106-2111, 2009/10/01 2009, doi: 10.1007/s11664-009-0861-9.

[4] B.-S. Lee, C.-W. Lee, and J.-W. Yoon, "Comparative study of Au–Sn and Sn–Ag–Cu as die-attach materials for power electronics applications," *Surface and Interface Analysis,* vol. 48, no. 7, pp. 493-497, 2016, doi: https://doi.org/10.1002/sia.5998.

[5] J. Xu, M. Wu, J. Pu, and S. Xue, "Novel Au-Based Solder Alloys: A Potential Answer for Electrical Packaging Problem," *Advances in Materials Science and Engineering,* vol. 2020, p. 4969647, 2020/05/26 2020, doi: 10.1155/2020/4969647.

[6] P. M. Struk and R. D. Pettegrew, "Soldering in Reduced Gravity Experiment," *SDTO 17003-U (SoRGE),* 2017.

[7] R. D. Pettegrew, P. M. Struk, J. K. Watson, D. R. Haylett, and R. S. Downs, "Gravitational effects on solder joints," *Welding journal,* vol. 82, no. 10, pp. 44-48, 2003.

[8] J. Easton, P. Struk, and A. Rotella, "Imaging and Analysis of Void-Defects in Solder Joints Formed in Reduced Gravity Using High-Resolution Computed Tomography," *46th Aerospace Sciences Meeting and Exhibit, Reno, NV;,* 2008.
[9] B. T. Lampe, "Room Temperature Aging Properties of Some Solder Alloys," *Welding Journal,* p. 11, 1976.
[10] N. Naden and T. J. Prater, "A Review of Welding in Space and Related Technologies," 2020.
[11] J. K. Watson, P. M. Struk, R. D. Pettegrew, and R. S. Downs, "Experimental investigation of solder joint defect formation and mitigation in reduced-gravity environments," *Journal of Spacecraft and Rockets,* vol. 44, no. 1, pp. 174-182, 2007.
[12] T. S. Balint, J. A. Cutts, E. A. Kolawa, and C. E. Peterson, "Extreme environment technologies for space and terrestrial applications," 2008, vol. 6960: SPIE, pp. 36-47.
[13] J. W. Sowards *et al.*, "Topical. Permanent Low-Earth Orbit Testbed for Welding and Joining: A Path Forward for the Commercialization of Space," 2022.
[14] P. M. Struk, J. W. Easton, G. P. Funk, G. S. Latta, A. W. Ganster, and B. E. Estes, "Recommendations for Enabling Manual Component Level Electronic Repair for Future Space Missions," 2011.
[15] R. Pettegrew, J. Easton, P. Struk, and E. Anderson, "In-flight manual electronics repair for deep-space missions," 2007: IEEE, pp. 1-16.
[16] P. M. Struk, R. C. Oeftering, and J. W. Easton, "Semi-Automated Diagnosis, Repair, and Rework of Spacecraft Electronics," in *AIAA Aerospace Sciences Meeting and Exhibit*, Reno, NV, 2008.
[17] R. C. Oeftering, M. A. Bradish, J. R. Juergens, M. J. Lewis, and D. R. Vrnak, "Component-Level Electronic-Assembly Repair (CLEAR) Operational Concept," in "NASA Technical Memorandum," 2011.
[18] R. C. Oeftering, M. A. Bradish, J. R. Juergens, M. J. Lewis, and D. R. Vrnak, "Component-Level Electronic-Assembly Repair (CLEAR) System Architecture," in "NASA Technical Memorandum," 2011.
[19] J. W. Easton and P. M. Struk, "Component Repair Experiment-1: An Experiment Evaluating Electronic Component-Level Repair During Spaceflight," 2012.
[20] C. D. Tudryn, "Solder joint fatigue study under low temperature Martian conditions," 2006: IEEE, pp. 9-pp.
[21] Q. Guan *et al.*, "Research Progress on the Solder Joint Reliability of Electronics Using in Deep Space Exploration," *Chinese Journal of Mechanical Engineering,* vol. 36, no. 1, pp. 1-13, 2023.
[22] B. Cornelius, S. Treivish, Y. Rosenthal, and M. Pecht, "The phenomenon of tin pest: A review," *Microelectronics Reliability,* vol. 79, pp. 175-192, 2017.
[23] R. N. Grugel, D. C. Gillies, F. Hua, and A. Anilkumar, "The In-Space Soldering Investigation: To Date Analysis of Experiments Conducted on the International Space Station," presented at the 2006 National Space and Missile Materials Symposium, Orlando, FL, June 26, 2006 - June 30, 2006, 2006.
[24] R. N. Grugel *et al.*, "Experiments Conducted Aboard the International Space Station: The Pore Formation and Mobility Investigation (PFMI) and the In-Space Soldering Investigation (ISSI): A Current Study of Results," presented at the 57th International Astronautical Congress (IAC) Conference, Valencia; Spain, 2006.

[25] J. W. Easton and P. M. STRUK, "Current Space Station Experiments Investigating Component Level Electronics Repair," in *Space 2009 Conference and Exhibit*, Pasadena, CA, September 14, 2009 - September 17, 2009 2009.
[26] J. W. Easton and P. M. Struk, "Soldering in a Reduced Gravity Environment (SoRGE)," in "NASA Technical Memorandum," 2012.
[27] K. A. Farley *et al.*, "Mars 2020 mission overview," *Space Science Reviews,* vol. 216, pp. 1-41, 2020.
[28] B. K. Muirhead, A. Nicholas, and J. Umland, "Mars sample return mission concept status," 2020: IEEE, pp. 1-8.
[29] R. M. Poorman, "Skylab M551 metals melting experiment," 1975.
[30] W. J. Ockels and L. Jagger-Meziere, "The Third ESA Student Parabolic-Flight Campaign," *ESA Bulletin,* vol. 105, pp. 91-100, 2001.
[31] G. L. Workman and W. F. Kaukler, "Laser welding in space," 1989.
[32] K. Kessel, "NEPP TRO," 2005.
[33] C. M. Miller, I. E. Anderson, and J. F. Smith, "A viable tin-lead solder substitute: Sn-Ag-Cu," *Journal of electronic materials,* vol. 23, pp. 595-601, 1994.
[34] I. E. Anderson, F. G. Yost, J. F. Smith, C. M. Miller, and R. L. Terpstra, "Pb-free Sn-Ag-Cu ternary eutectic solder," ed: Google Patents, 1996.
[35] J. K. Watson, P. M. Struk, R. D. Pettegrew, and R. S. Downs, "Experimental Investigation of Solder Joint Defect Formation and Mitigation in Reduced-Gravity Environments," *AIAA Journal of Spacecraft and Rockets,* vol. 44, no. 1, pp. 174-182, 2007.
[36] T. Carlberg and M. Liljendahl, "Soldering Under Microgravity," in *Proceedings of the 4th European Symposium on Materials Sciences under Microgravity, ESA SP-191*, 1983, pp. 337 – 342.
[37] C. A. Winter, *The Microgravity Research Experiments (MICREX) data base [microform] / C.A. Winter, J.C. Jones* (NASA technical memorandum ; 108523., no. Accessed from https://nla.gov.au/nla.cat-vn4146855). MSFC, Ala. : [Springfield, Va: National Aeronautics and Space Administration, Marshall Space Flight Center ; National Technical Information Service, distributor, 1996.
[38] B. Thomas, A. Atkinson, and R. J. Dashwood, "Strength of Soldered Joints Formed under Microgravity Conditions," *Journal of Electronic Materials,* vol. 36, no. 1, pp. 1-5, 2007/01/01 2007, doi: 10.1007/s11664-006-0033-0.
[39] A. Dunkle, S. Itschner, and J. Kuhlman, "Magnetic solder to reduce porosity of solder joints formed in microgravity," *Acta Astronautica,* vol. 178, pp. 463-470, 01/01 2021, doi: 10.1016/j.actaastro.2020.09.033.
[40] R. Grugel, "Final Research Report: In-Space Soldering Investigation (ISSI), City," ed, 2006.
[41] R. N. Grugel *et al.*, "Materials research conducted aboard the International Space Station: Facilities overview, operational procedures, and experimental outcomes," *Acta Astronautica,* vol. 62, no. 8-9, pp. 491-498, 2008.
[42] J. F. Zhao, G. Liu, S. X. Wan, and N. Yan, "Bubble dynamics in nucleate pool boiling on thin wires in microgravity," *Microgravity Science and Technology,* vol. 20, pp. 81-89, 2008.
[43] N. O. Young, J. S. Goldstein, and M. J. Block, "The motion of bubbles in a vertical temperature gradient," *Journal of Fluid Mechanics,* vol. 6, no. 3, pp. 350-356, 1959.

[44] S. B. R. Karri, "Experimental study of simulated micro-gravity vapor-liquid flow regimes," Ph.D., University of New Hampshire, United States -- New Hampshire, 8816692, 1988. [Online]. Available: https://www.proquest.com/dissertations-theses/experimental-study-simulated-micro-gravity-vapor/docview/303693436/se-2?accountid=10906[Online]. Available: https://quicksearch.lib.iastate.edu/openurl/01IASU_INST/01IASU_INST:01IASU?url_ver=Z39.88-2004&rft_val_fmt=info:ofi/fmt:kev:mtx:dissertation&genre=dissertations&sid=ProQ:ProQuest+Dissertations+%26+Theses+Global&atitle=&title=Experimental+study+of+simulated+micro-gravity+vapor-liquid+flow+regimes&issn=&date=1988-01-01&volume=&issue=&spage=&au=Karri%2C+S.+B.+Reddy&isbn=979-8-207-09308-6&jtitle=&btitle=&rft_id=info:eric/&rft_id=info:doi/

[45] S. M. O'Shaughnessy and A. J. Robinson, "Numerical investigation of bubble induced Marangoni convection: some aspects of bubble geometry," *Microgravity Science and Technology,* vol. 20, pp. 319-325, 2008.

[46] H. Yin and T. Emi, "Marangoni flow at the gas/melt interface of steel," *Metallurgical and Materials Transactions B,* vol. 34, pp. 483-493, 2003.

[47] G. Smith, R. Spievy, P. Luz, R. Grugel, A. Anikumar, and M. Volz, "Toward understanding pore formation and mobility during controlled directional solidification in a microgravity environment investigation (PFMI)," 2013, p. 5119.

[48] M. E. Glicksman, M. B. Koss, and E. A. Winsa, "Dendritic growth velocities in microgravity," *Physical review letters,* vol. 73, no. 4, p. 573, 1994.

[49] O. Funke *et al.*, "Dendrite growth velocity in levitated undercooled nickel melts," *Journal of Crystal Growth,* vol. 297, no. 1, pp. 211-222, 2006/12/15/ 2006, doi: https://doi.org/10.1016/j.jcrysgro.2006.08.045.

[50] J. Gao, M. Han, A. Kao, K. Pericleous, D. V. Alexandrov, and P. K. Galenko, "Dendritic growth velocities in an undercooled melt of pure nickel under static magnetic fields: A test of theory with convection," *Acta Materialia,* vol. 103, pp. 184-191, 2016.

[51] S. Ozawa, J. Yoshizawa, T. Saito, and T. Motegi, "Microstructure of Nd–Fe–B Alloys Solidified under Microgravity Conditions," *Materials Transactions, JIM,* vol. 41, no. 9, pp. 1121-1124, 2000.

[52] S. A. Nabavizadeh, M. Eshraghi, S. D. Felicelli, S. N. Tewari, and R. N. Grugel, "The Marangoni convection effects on directional dendritic solidification," *Heat and Mass Transfer,* vol. 56, pp. 1329-1341, 2020.

[53] S. A. Nabavizadeh, M. Eshraghi, S. D. Felicelli, S. N. Tewari, and R. N. Grugel, "Effect of bubble-induced Marangoni convection on dendritic solidification," *International Journal of Multiphase Flow,* vol. 116, pp. 137-152, 2019.

[54] C. Reynard, M. Barthes, R. Santini, and L. Tadrist, "Experimental study of the onset of the 3D oscillatory thermocapillary convection around a single air or vapor bubble.: Influence on heat transfer," *Experimental thermal and fluid science,* vol. 29, no. 7, pp. 783-793, 2005.

[55] S. Barsi, M. Kassemi, and J. I. D. Alexander, "Effects of void-induced convection on interface morphology and segregation during low-g solidification," *International journal of heat and mass transfer,* vol. 47, no. 23, pp. 5129-5137, 2004.

[56] M. Kassemi, S. Barsi, M. Kaforey, and D. Matthiesen, "Effect of void location on segregation patterns in microgravity solidification," *Journal of Crystal Growth,* vol. 225,

no. 2, pp. 516-521, 2001/05/01/ 2001, doi: https://doi.org/10.1016/S0022-0248(01)00945-9.

[57] S. Ghosh, B. Roy, S. Paul, S. Das, S. Roy, and A. Banerjee, "Writing with lasers: a new technique of controlled lithography using thermooptically manipulated microbubbles," 2015, vol. 9654: SPIE, pp. 14-21.

[58] Y. Z. Li, N. Mangelinck-Noël, G. Zimmermann, L. Sturz, and H. Nguyen-Thi, "Effect of solidification conditions and surface pores on the microstructure and columnar-to-equiaxed transition in solidification under microgravity," *Journal of Alloys and Compounds,* vol. 749, pp. 344-354, 2018.

[59] S. C. Hardy, "The motion of bubbles in a vertical temperature gradient," *Journal of Colloid and Interface Science,* vol. 69, no. 1, pp. 157-162, 1979/03/15/ 1979, doi: https://doi.org/10.1016/0021-9797(79)90090-0.

[60] W. Dreyer and W. H. Müller, "A study of the coarsening in tin/lead solders," *International Journal of Solids and Structures,* vol. 37, no. 28, pp. 3841-3871, 2000/07/01/ 2000, doi: https://doi.org/10.1016/S0020-7683(99)00146-8.

[61] K. Nogita *et al.*, "XRD study of the kinetics of β↔ α transformations in tin," *Philosophical Magazine,* vol. 93, no. 27, pp. 3627-3647, 2013.

[62] S. R. Kalidindi and S. Pathak, "Determination of the effective zero-point and the extraction of spherical nanoindentation stress–strain curves," *Acta Materialia,* vol. 56, no. 14, pp. 3523-3532, 2008, doi: http://dx.doi.org/10.1016/j.actamat.2008.03.036.

[63] S. Pathak and S. R. Kalidindi, "Spherical nanoindentation stress–strain curves," *Materials Science and Engineering: R: Reports,* vol. 91, no. 0, pp. 1-36, 5// 2015, doi: http://dx.doi.org/10.1016/j.mser.2015.02.001.

[64] J. S. Weaver, A. Khosravani, A. Castillo, and S. R. Kalidindi, "High throughput exploration of process-property linkages in Al-6061 using instrumented spherical microindentation and microstructurally graded samples," *Integrating Materials and Manufacturing Innovation,* vol. 5, pp. 192-211, 2016.

[65] M. A. Meyers and E. Ashworth, "A model for the effect of grain size on the yield stress of metals," *Philosophical Magazine A (Physics of Condensed Matter, Defects and Mechanical Properties),* vol. 46, no. Copyright 1983, IEE, pp. 737-59, 1982.

[66] S. Pathak, J. Michler, K. Wasmer, and S. R. Kalidindi, "Studying grain boundary regions in polycrystalline materials using spherical nano-indentation and orientation imaging microscopy," *Journal of Materials Science,* vol. 47, no. 2, pp. 815-23, 2012, doi: 10.1007/s10853-011-5859-z.

IOWA STATE UNIVERSITY

Structure and Properties of the Terrestrial vs. Microgravity solders under extreme conditions of Elevated and Cryo Temperatures

Manish Kumar, Ph.D. Candidate
Ralph Napolitano, Professor
Sid Pathak, Assistant Professor

Department of Materials Science and Engineering

Acknowledgements:
NASA PSI (#80NSSC20K0223 – FY2020)
NASA PSI (#80NSSC23K0279 – FY2023)
NASA EPSCoR(#80NSSC21M0229)
Dr. Louise L Littles (NASA Marshall Space Flight Center)
Dr. Binayak Panda (NASA Marshall Space Flight Center)

Motivation: In-Space Electronics Repair and Metal Joining for Space Missions

Roadmaps for future space research, ESA
NASA PSI, ASGSR 2022

Exploration
- Lunar construction/In-Space resource utilization
- In-Space manufacturing and repair
- Integrated computation materials engineering
- Thermophysical properties

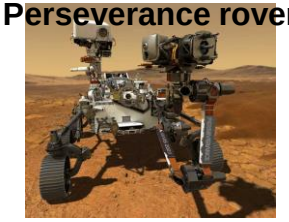
In-Space Manufacturing and Repair
- Additive manufacturing
- Capabilities to fabricate spare parts
- Electronic components repair and metal joining
- Glass forming Techniques
- In-Space sintering

In-Space Metal joining
- In-Space Soldering
- In-Space welding
- In-Space Brazing

Soldering capabilities in space to enable space-based fabrication and repairs

- Essential for future human exploration missions to ISS and beyond low earth orbit
- NASA Mars Rover sample return mission
- Manned missions to Moon and Mars

Perseverance rover



Soldering in microgravity vs. gravity

- In terrestrial conditions: Heat and mass transfer - affected by gravity-induced natural convection
- In microgravity conditions:
  - Reduced fluid flow in the absence of gravity
  - No buoyancy and sedimentation
  - Removal of buoyancy convection enables the study of thermocapillary effects

Large temperature variations during space exploration missions

- Moon: +120°C to -230°C
- Outside ISS: +121°C to -157°C
- Mars: +20°C to -153°C

➢ Moon/Mars rover can not function well under extreme temperatures
  - Warm Electronic Box (WEB) - protect rover's "vital organs" – maintain temperature between +40°C to -40°C
  - Distributed, non-heated architecture for future missions

➢ Inadequate reliability of solder joints for the extreme cryo and elevated temperatures

Courtesy: Dr. Fran Chiaramonte, Program Scientist NASA, Status of CLD and Decadal survey
Roadmaps for future space research, ESA

Grugel, et al. Final Research Report. In-Space Soldering Investigation (ISSI). 11-51. 2011
C. D. Tudryn, "Solder joint fatigue study under low temperature Martian conditions." pp. 9-pp

In-Space Soldering investigation (2003-04): Gravity independent thermocapillary

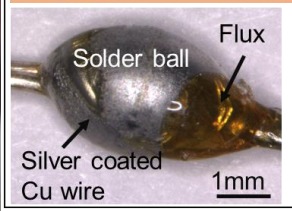
Wire wrap method

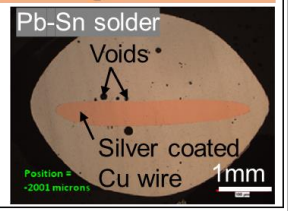


ISS Science Officer, Mike Fincke, conducting the soldering experiments

Near Eutectic 40wt%Pb-60wt%Sn alloy(Kester "44"® 60-40 rosin core solder (3.3 wt.% rosin)

Wire Wrap Solder Method
- Fewer porosity
- Smaller pore size



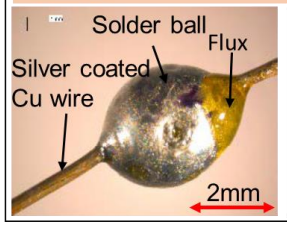


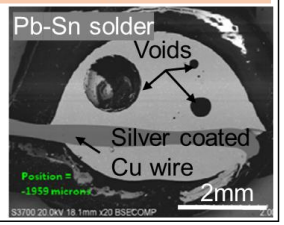
- Thermocapillary flow
  - Gravity-independent flux rotation
  - Flux migrates to the hot end of the sample and freezes

Wire Feed method



Wire Feed Solder Method
- Increased surface defect
- Increased pore size





- Solder wire was fed on a heated Cu-wire (Silver plated)
- More flux trapped internally

- **Limited control** over soldering parameters such as **Temperature**, **hold time at max temperature**, **heating and cooling rate, soldering iron contact**
- Planning experiments with precise control of these parameters

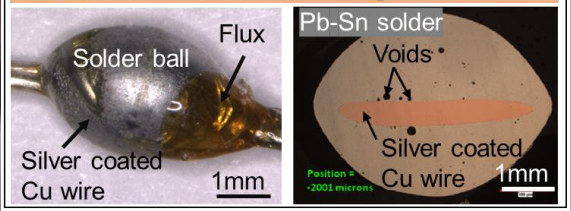
In-Space Soldering investigation (2003-04): Gravity independent thermocapillary

Wire wrap method


ISS Science Officer, Mike Fincke, conducting the soldering experiments

Near Eutectic 40wt%Pb-60wt%Sn alloy(Kester "44"® 60-40 rosin core solder (3.3 wt.% rosin)

Wire Wrap Solder Method • Fewer porosity • Smaller pore size

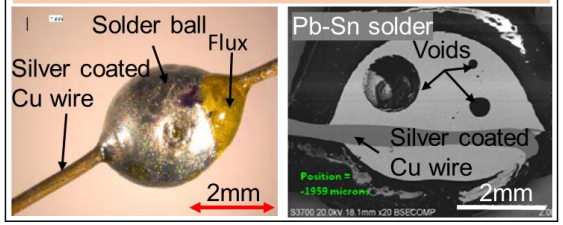


- Thermocapillary flow
  - Gravity-independent flux rotation
  - Flux migrates to the hot end of the sample and freezes

Wire Feed method



Wire Feed Solder Method • Increased surface defect • Increased pore size

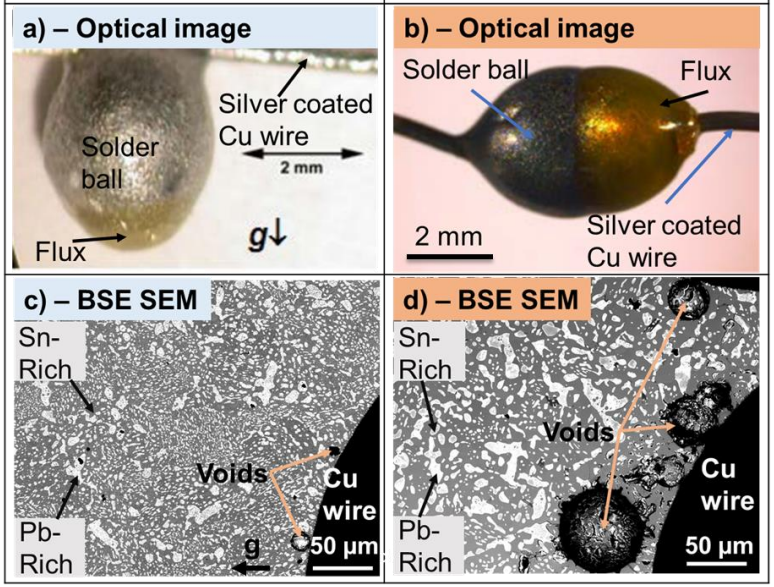


- Solder wire was fed on a heated Cu-wire (Silver plated)
- More flux trapped internally

- **Limited control** over soldering parameters such as **Temperature**, **hold time at max temperature**, **heating and cooling rate, soldering iron contact**
- Planning experiments with precise control of these parameters

Based on wire wrap 40Pb-60Sn solder sample - differences in soldering in microgravity vs. terrestrial conditions:

Solder prepared in terrestrial conditions	Solder prepared in microgravity conditions
a) – Optical image	b) – Optical image
c) – BSE SEM	d) – BSE SEM



- ❑Dominating forces due to surface tension – leading to football shape of microgravity solder.
- ❑Increased porosity – affect the mechanical integrity, thermal and electrical conductivity.
- ❑Effect of composition on void formation in microgravity is unknown; other compositions are:
  - Off-eutectic 50Pb-50Sn
  - Lead-free Sn-Ag-Cu

Main Scientific Questions

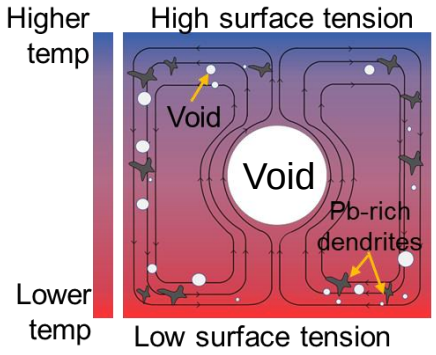
i) What factors contribute to accumulation of voids (increased number and larger voids) in microgravity solder?

ii) What are potential void mitigation strategies?

iii) How do extreme temperature conditions (elevated and cryogenic) affect the solder properties?

Objective 1: Flight experiment 1: Characterization of the solder-joint from the ISSI (2003-04)

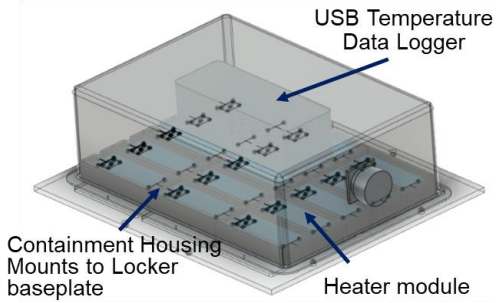
- Effect of lack of buoyancy and natural convection on void formation in microgravity solder
- Effect of thermal convection on solder microstructure in microgravity
- Effect of void-induced Marangoni flow on solder structure



Objective 2: Flight experiment 2: New microgravity and terrestrial soldering experiments with precise control of soldering parameters (Summer 2024)

Parameters/Factors to be controlled:

- Solder composition
  - 40Pb-60Sn
  - 50Pb-50Sn
  - Sn-Ag-Cu
- Solder substrate – Cu and gold
- Thermal profile
- Flux type



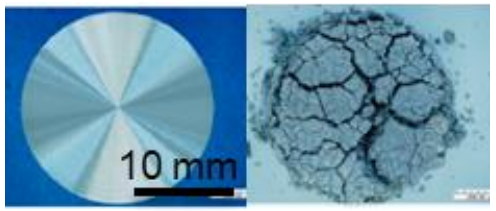
Objective 3: Investigating the effects of extreme temperature conditions of space on solder joint properties

Elevated temperature:

- Effect of room and elevated temperature aging on solder properties

Cryogenic temperature:

- Study the effect of β-Sn to α-Sn transformation kinetics on solder properties



Main Scientific Questions

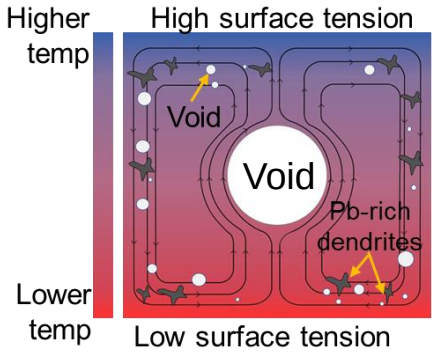
i) What factors contribute to accumulation of voids (increased number and larger voids) in microgravity solder?

ii) What are potential void mitigation strategies?

iii) How do extreme temperature conditions (elevated and cryogenic) affect the solder properties?

Objective 1: Flight experiment 1: Characterization of the solder-joint from the ISSI (2003-04)

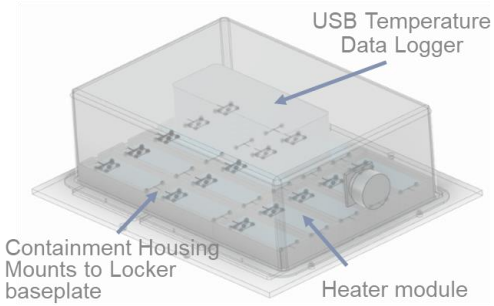
- Effect of lack of buoyancy and natural convection on void formation in microgravity solder
- Effect of thermal convection on solder microstructure in microgravity
- Effect of void-induced Marangoni flow on solder structure



Objective 2: Flight experiment 2: New microgravity and terrestrial soldering experiments with precise control of soldering parameters (Summer 2024)

Parameters/Factors to be controlled:

- Solder composition
  - 40Pb-60Sn
  - 50Pb-50Sn
  - Sn-Ag-Cu
- Solder substrate – Cu and gold
- Thermal profile
- Flux type



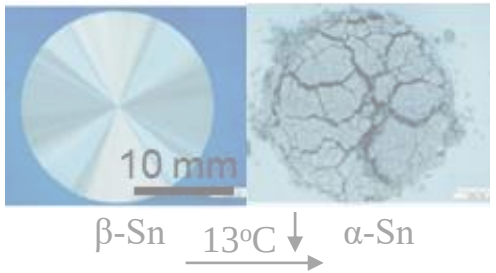
Objective 3: Investigating the effects of extreme temperature conditions of space on solder joint properties

Elevated temperature:

- Effect of room and elevated temperature aging on solder properties

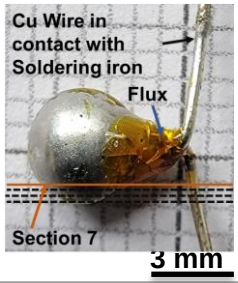
Cryogenic temperature:

- Study the effect of β-Sn to α-Sn transformation kinetics on solder properties

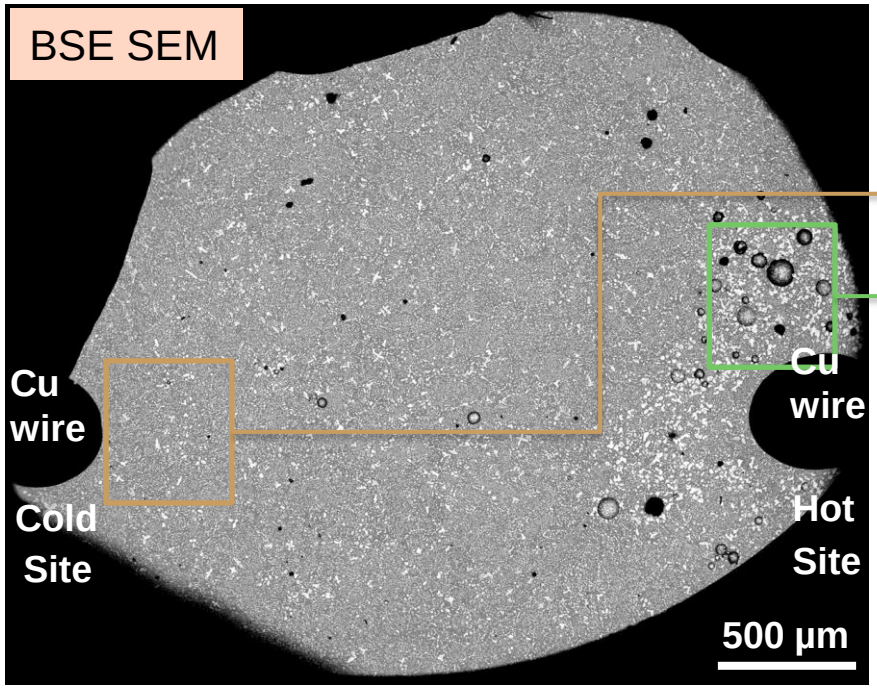


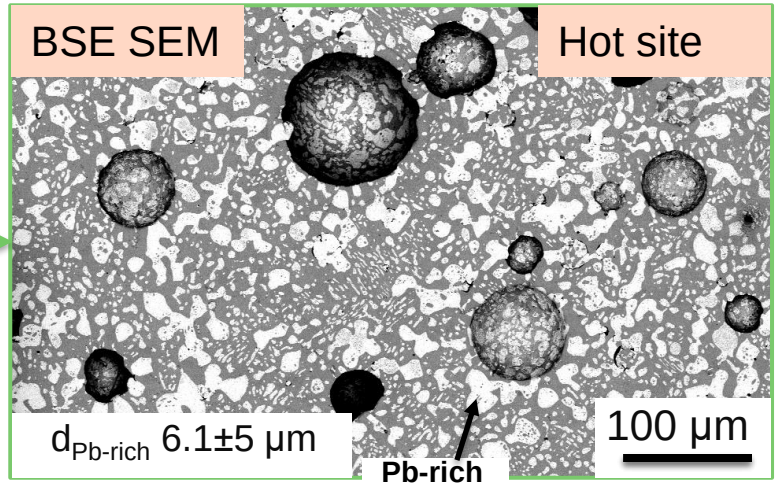
Inhomogeneous microstructure of ISSI Microgravity Wire Feed solder

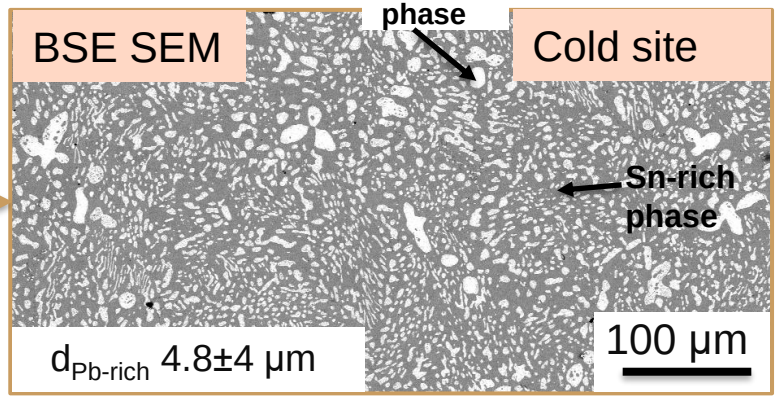
ISSI Microgravity Wire Feed Small loop solder – section 7 (1.2 mm away from solder edge)











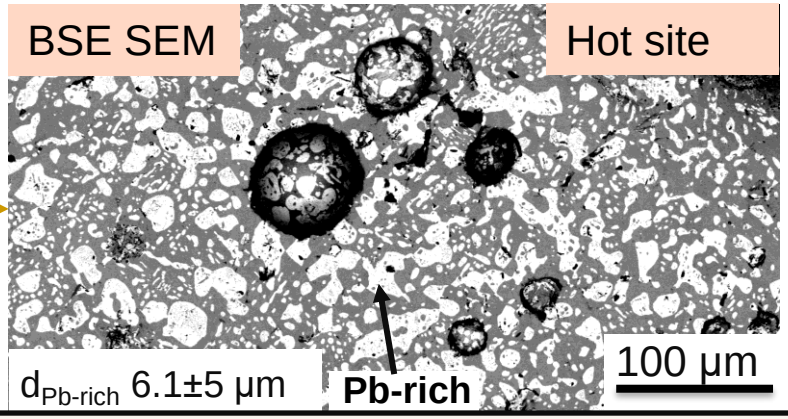
- Inhomogeneous microstructure
- Accumulation of voids and Pb-Rich
- Similar observations - across 10 different cross-sections

Inhomogeneous microstructure of ISSI Microgravity Wire Feed solder

ISSI Microgravity Wire Feed Small loop solder – section 7 (1.2 mm away from solder edge)



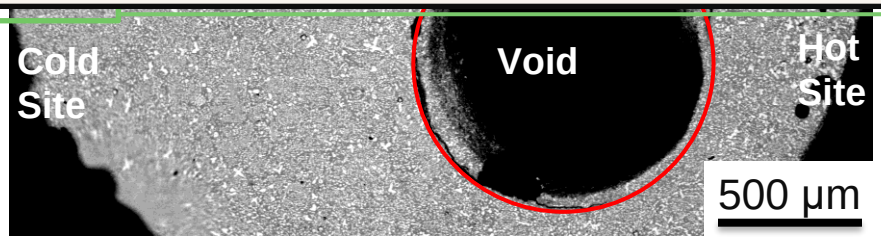




What factors contribute to this void accumulation?

Factors due to temperature gradient:

- Marangoni force
- Thermal convection
- Void-induced Marangoni convection



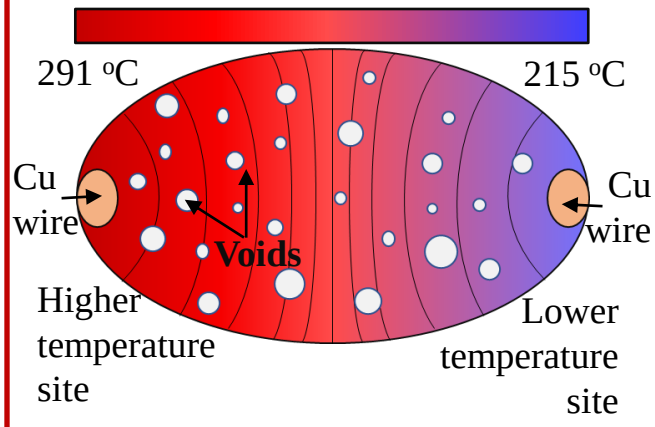
- Accumulation of voids and Pb-Rich
- Higher eutectic spacing toward hot site – slow cooling rate
- Higher amount of primary Pb-rich dendrites (32%) in hot region than the cold region (9%)

Parameter	Hot site	Cold site
Eutectic Spacing (µm)	2.15 ± 0.45	1.89 ± 0.43
Primary Pb-rich dendrites (%)	32	9
Corresponding composition	52Pb-48Sn	42Pb-58Sn

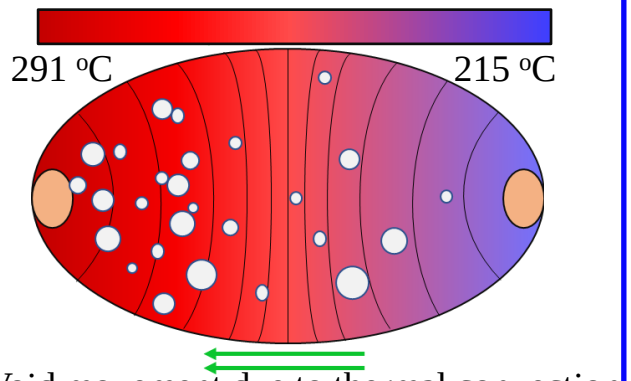
Thermal convection and Marangoni flow causes void accumulation

Effect of thermal convection on void accumulation

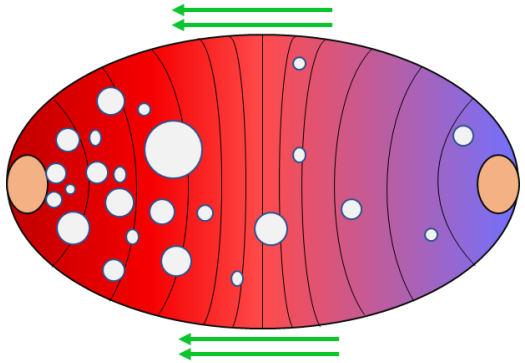
Time step 1) Randomly dispersed voids just after melting



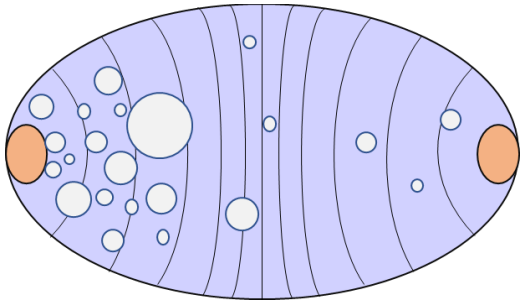
Time step 2) Void movement toward higher temperature site



Time step 3) Void accumulation/coalescence



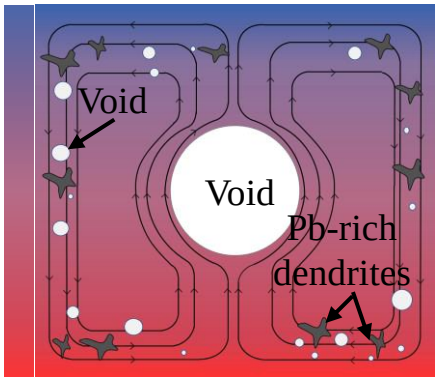
Time step 4) Accumulated voids at the higher temperature site after Solidification



Effect of Marangoni Flow on void and dendrites accumulation

a) Marangoni Flow in Molten state

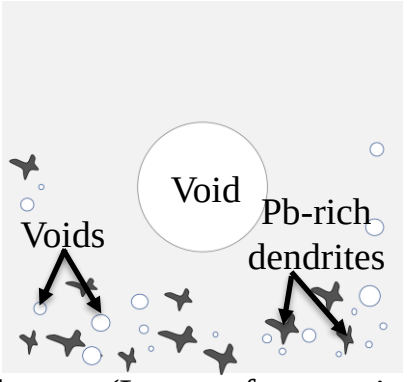
Low Temp (High surface tension)



High temp (Low surface tension)

b) Accumulated voids and dendrites (toward the higher temp region) after solidification

Low Temp (High surface tension)



High temp (Low surface tension)

- Thermal convection and Marangoni flow probably causes the void and primary dendrites accumulation

Ghosh, Banerjee, 2015. Writing with lasers: a new technique of controlled lithography using thermooptically manipulated microbubbles. (Vol. 9654, pp. 14-21). SPIE.
Nabavizadeh, Eshraghi. International Journal of Multiphase Flow 2019, Effect of bubble-induced Marangoni convection on dendritic solidification
Yuan. MMT A 2019. Marangoni-Convection-Driven Bubble Behavior and Microstructural Evolution of SnAgSnBiCu in microgravity

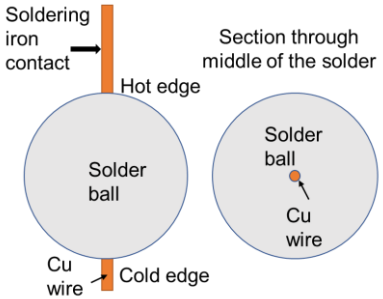
Marangoni flow dominating over diffusive transport

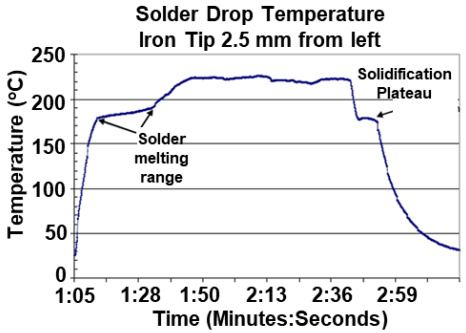
Assumption: heat transfer only via conduction

Average heat transfer $Q(J) = mC_p(T_P - T_m) + mh_f$

Heat of fusion

Heat transfer via conduction $= k_{Cu}\, A_{Cu} \frac{dT}{dx}$





Temperature profile from ISSI terrestrial experiments

Temperature difference from the hot end of the solder ball (3.8 mm) to the cold end = 76°C

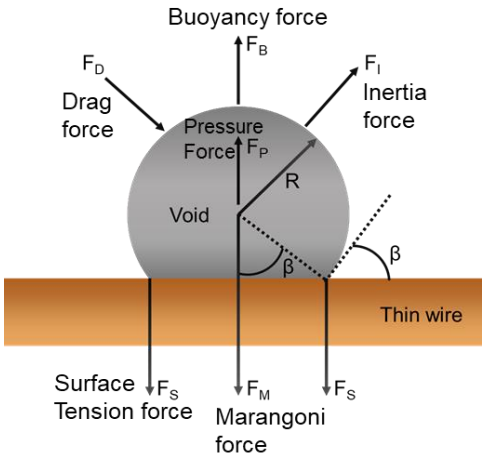
A distinct surface tension gradient with low surface tension in the high-temperature region and high surface tension in the low-temperature region.

Marangoni number (Ma): $\frac{Marangoni\ transport\ rate}{Diffusive\ transport\ rate} = -\frac{d\sigma}{dT}\left(\frac{L \times dT}{\mu \times \alpha}\right)$ = 223

Reynolds number (Re): $\left(\frac{\rho \times u \times L}{\mu}\right)$ = 1832

L = characteristic length u = Flow speed μ = dynamic viscosity

α = thermal diffusivity dT = temperature gradient

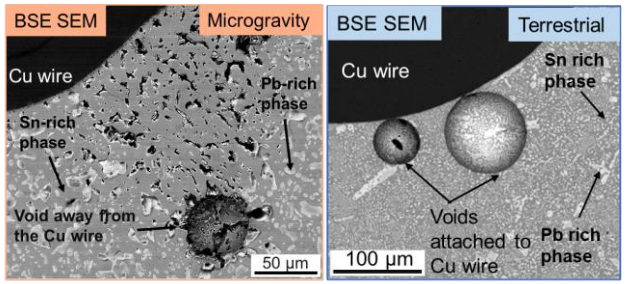


Accumulated voids and dendrites (toward the higher temp region) – probably caused by strong Marangoni flow demonstrated by higher Marangoni number

Grugel, et al. Final Research Report. In-Space Soldering Investigation (ISSI). 11-51. 2011
Watson et al., "Experimental Investigation of Solder Joint Defect Formation and Mitigation in Reduced-Gravity Environments," AIAA Journal of Spacecraft and Rockets, vol. 44, no. 1, pp. 174-182, 2007.
N. O. Young, J. S. Goldstein, and M. J. Block, "The motion of bubbles in a vertical temperature gradient," Journal of Fluid Mechanics, vol. 6, no. 3, pp. 350-356, 1959

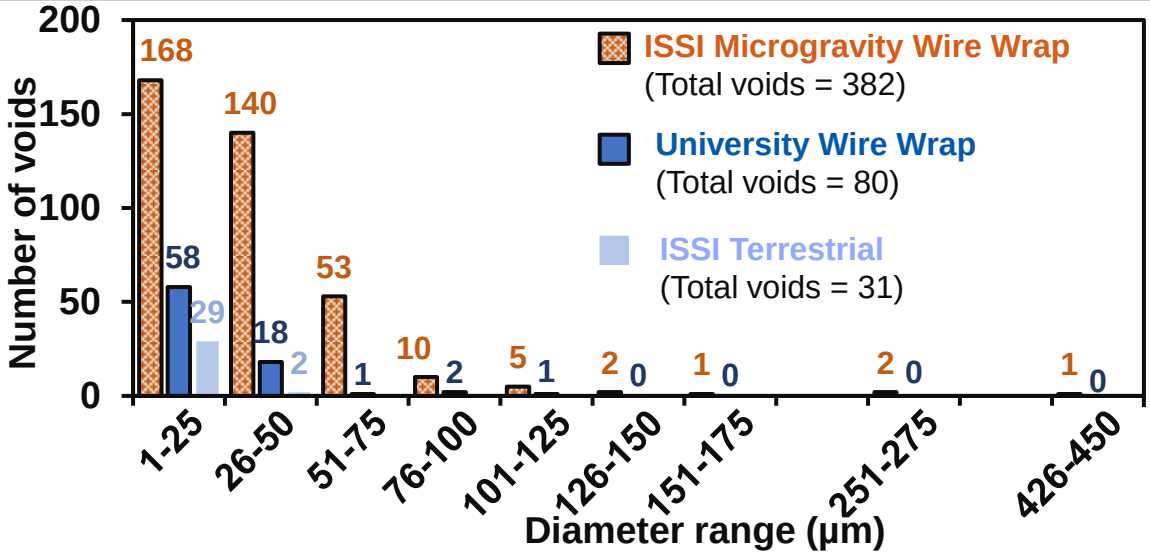
Obj 1. ISS Flight #1 - ISSI (2003-04) solder characterization

Increased porosity in ISSI Microgravity solder than Terrestrial solder

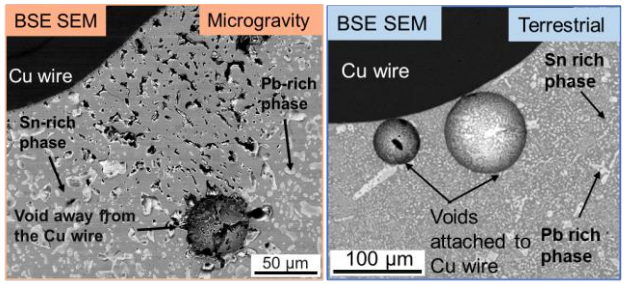


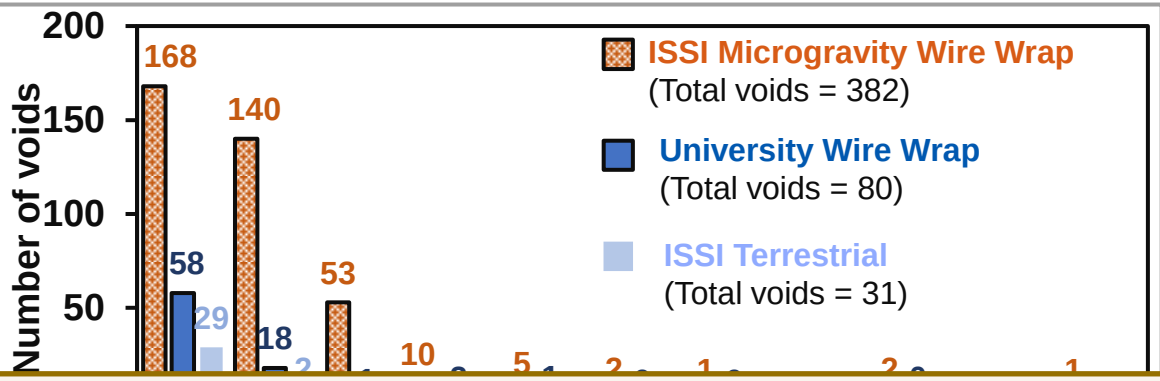
g

- Higher void volume fraction in microgravity solder - due to absence of buoyancy force and slower fluid mixing

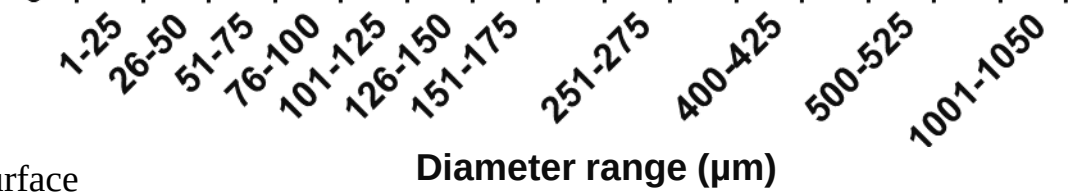


Increased porosity in ISSI Microgravity solder than Terrestrial solder





- Higher void volume fraction in wire feed solder than wire wrap solder
- In **wire wrap solder**: solder is free to melt around the Cu wire - any flux in the core can exit and remain on the molten solder surface



- In **wire feed solder**: after a small amount of liquid solder accrues - any flux/rosin from the core is now constrained to remain internal to the liquid

Summary of objective 1:

- Lack of Buoyancy and natural convection causes excess voids in microgravity solder than terrestrial solder.
- Thermal convection due to density gradient and void-induced Marangoni convection possibly affect the void accumulation and solder microstructure in microgravity.
- Uncontrolled soldering experiments may have affected the solder microstructure and void formation during the ISSI.

Main Scientific Questions

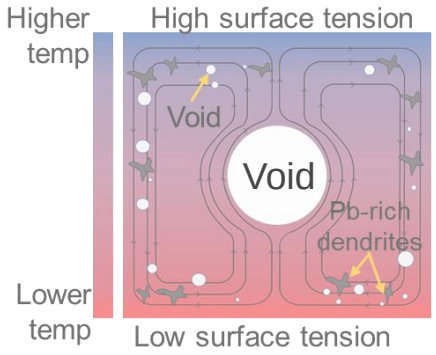
i) What factors contribute to accumulation of voids (increased number and larger voids) in microgravity solder?

ii) What are potential void mitigation strategies?

iii) How do extreme temperature conditions (elevated and cryogenic) affect the solder properties?

Objective 1: Flight experiment 1: Characterization of the solder-joint from the ISSI (2003-04)

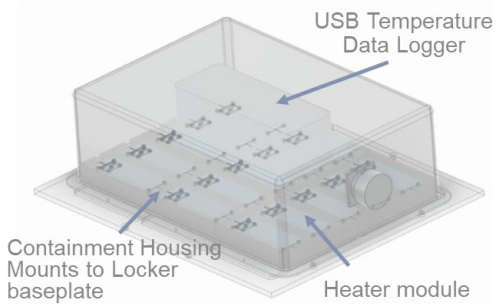
- Effect of lack of buoyancy and natural convection on void formation in microgravity solder
- Effect of thermal convection on solder microstructure in microgravity
- Effect of void-induced Marangoni flow on solder structure



Objective 2: Flight experiment 2: New microgravity and terrestrial soldering experiments with precise control of soldering parameters (Summer 2024)

Parameters/Factors to be controlled:

- Solder composition
  - 40Pb-60Sn
  - 50Pb-50Sn
  - Sn-Ag-Cu
- Solder substrate – Cu and gold
- Thermal profile
- Flux type



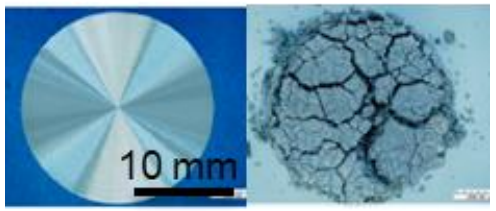
Objective 3: Investigating the effects of extreme temperature conditions of space on solder joint properties

Elevated temperature:

- Effect of room and elevated temperature aging on solder properties

Cryogenic temperature:

- Study the effect of β-Sn to α-Sn transformation kinetics on solder properties



β-Sn $13^{o}C$ ↓ α-Sn

Obj 3 Effect of elevated and cryo temperature on solder properties

Solder joints under extreme Elevated and Cryo temperature conditions

- ISS experiences extreme elevated (+121°C) and cryo (-157°C) temperature 16 times in a day
- Use of Warm Electronics Box (WEB) in rovers- maintain temperature between +40°C to -40°C
- Future space missions: a distributed, non-heated architecture for the rover's electronics
- Components will be placed near the systems they interact with



Warm Electronics Box (WEB)

Solder joints at RT/Elevated temperature

- Room temperature for 40Pb-60Sn and 50Pb-50Sn solder is ~ 0.65Tm
- Room temperature for 95.5Sn-3.8Ag-0.7Cu ~ 0.61Tm

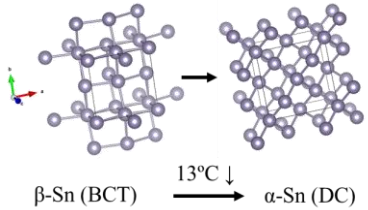
Grain coarsening due to

- Classical diffusion – initiate the grain coarsening
- Thermo-mechanical stress – helps in coarsening
- Surface tension – control the shape of the coarsened grain

- Aging may influence the performance and reliability of the solder joints for In-Space application

Solder joints at cryogenic temperature

- In Sn-based solders
  - Allotropic β-Sn to α-Sn transformation below 13°C



  - β → α transformation – Nucleation and growth process
  - 26% increase in volume
- Ductile to brittle transition in 40Pb-60Sn solder: -80°C
- DBT in Sn-Ag-Cu solder: -70°C

C. D. Tudryn, "Solder joint fatigue study under low temperature Martian conditions." pp. 9-pp
Q. Guan, "Research Progress on the Solder Joint Reliability in Deep Space Exploration," vol. 36, no. 1, pp. 1-13, 2023
B. Cornelius., "The phenomenon of tin pest: A review," Microelectronics Reliability, vol. 79, pp. 175-192, 2017

Obj 3 Effect of elevated and cryo temperature on solder properties

Solder joints under extreme Elevated and Cryo temperature conditions

- ISS experiences extreme elevated (+121°C) and cryo (-157°C) temperature 16 times in a day
- Use of Warm Electronics Box (WEB) in rovers- maintain temperature between +40°C to -40°C
- Future space missions: a distributed, non-heated architecture for the rover's electronics
- Components will be placed near the systems they interact with



Warm Electronics Box (WEB)

Solder joints at RT/Elevated temperature

- Room temperature for 40Pb-60Sn and 50Pb-50Sn solder is ~ 0.65Tm
- Room temperature for 95.5Sn-3.8Ag-0.7Cu ~ 0.61Tm

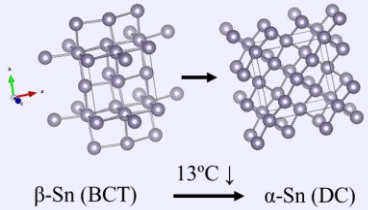
Grain coarsening due to

- Classical diffusion – initiate the grain coarsening
- Thermo-mechanical stress – helps in coarsening
- Surface tension – control the shape of the coarsened grain

- Aging may influence the performance and reliability of the solder joints for In-Space application

Solder joints at cryogenic temperature

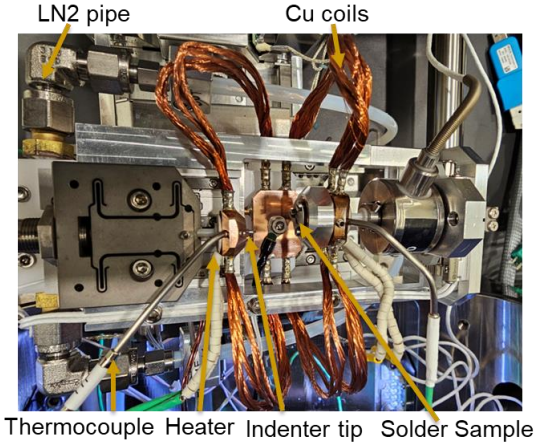
- In Sn-based solders
  - Allotropic β-Sn to α-Sn transformation below 13°C



  - β → α transformation – Nucleation and growth process
  - 26% increase in volume
- Ductile to brittle transition in 40Pb-60Sn solder: -80°C
- DBT in Sn-Ag-Cu solder: -70°C

C. D. Tudryn, "Solder joint fatigue study under low temperature Martian conditions." pp. 9-pp
Q. Guan, "Research Progress on the Solder Joint Reliability in Deep Space Exploration," vol. 36, no. 1, pp. 1-13, 2023
B. Cornelius., "The phenomenon of tin pest: A review," Microelectronics Reliability, vol. 79, pp. 175-192, 2017

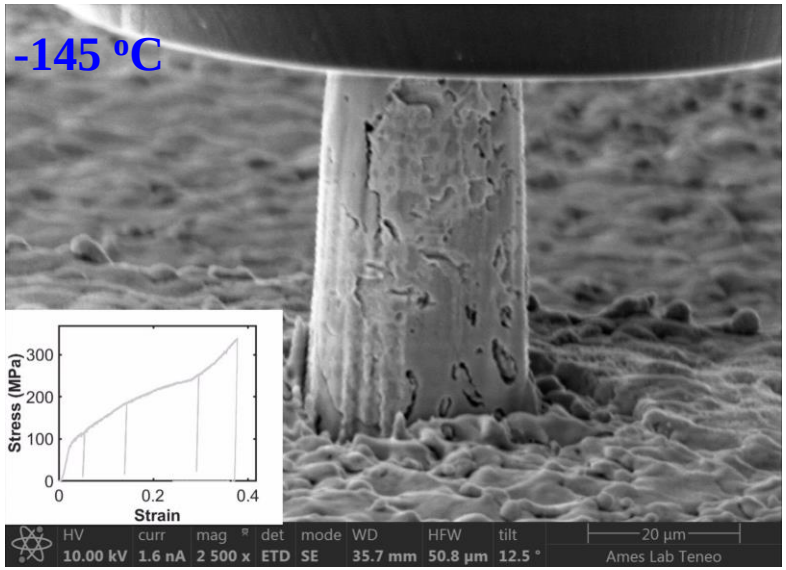
Increased strength of Terrestrial 40Pb-60Sn at cryogenic temperatures



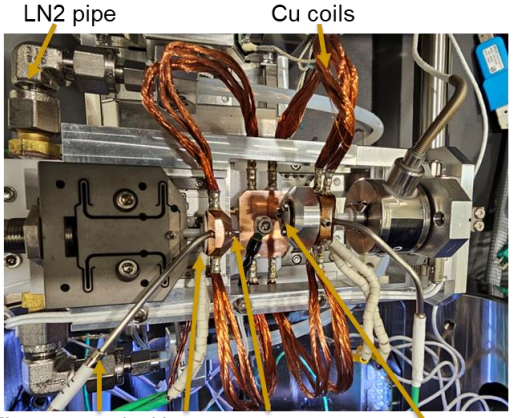
Alemnis In-Situ SEM nanoindenter

Order of the experiments:

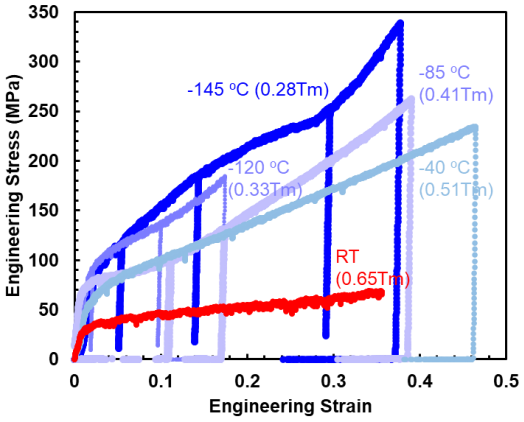
1) Room temperature (RT) 2) -145 ºC
3) -120 ºC 4) -85ºC 5) -40ºC 6) RT

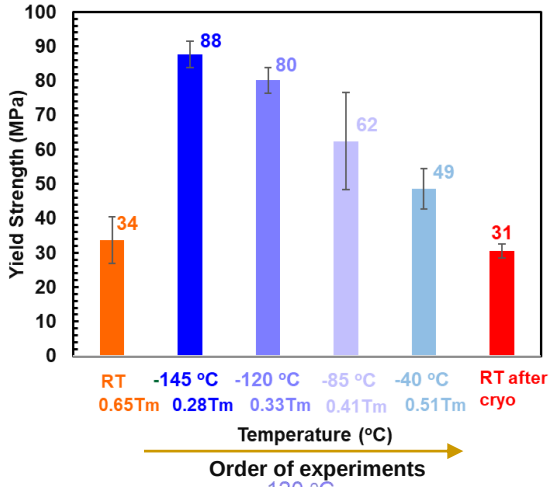


Increased strength of Terrestrial 40Pb-60Sn at cryogenic temperatures



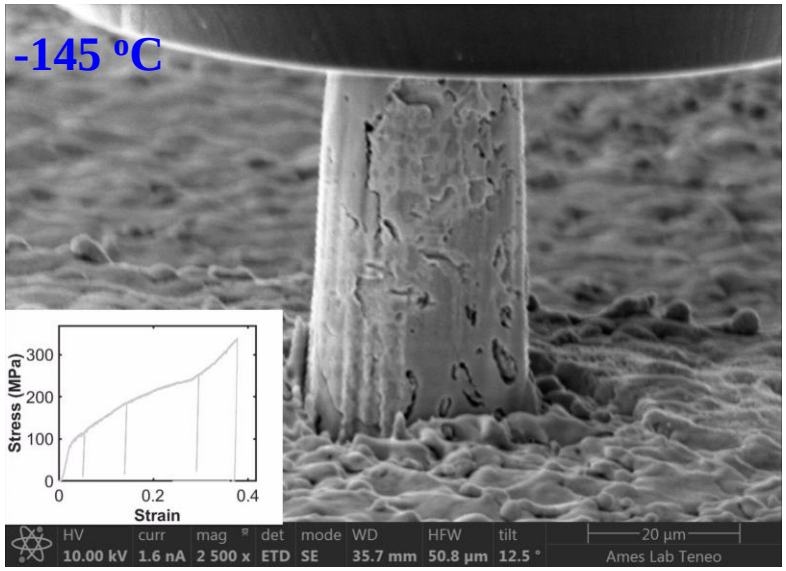
Alemnis In-Situ SEM nanoindenter

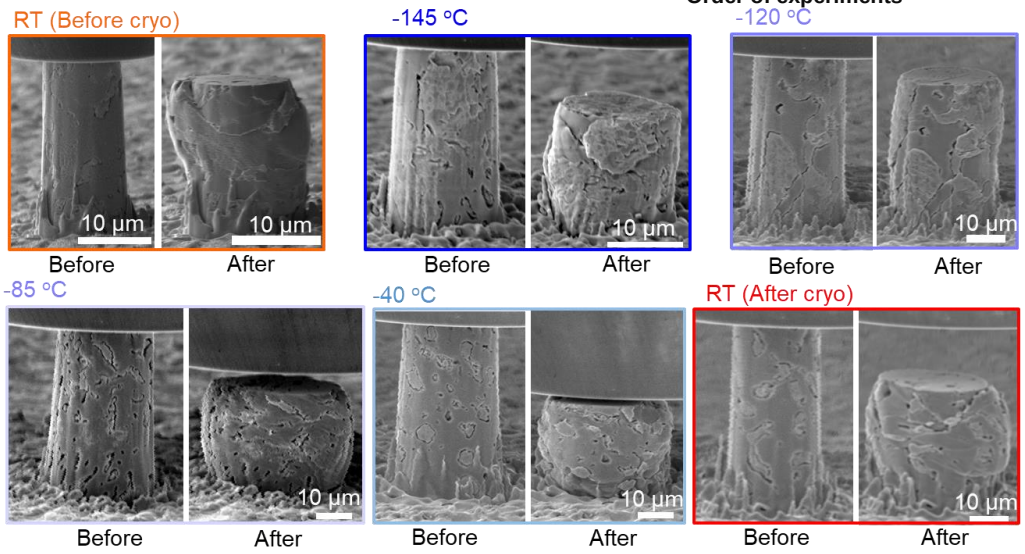




Order of the experiments:

1) Room temperature (RT) 2) -145 °C
3) -120 °C 4) -85°C 5) -40°C 6) RT

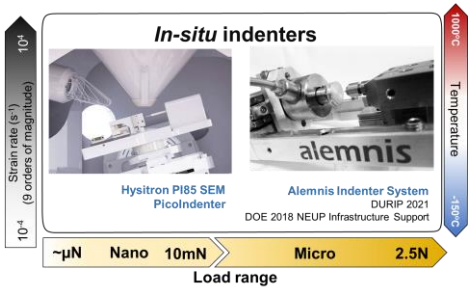




- Solder demonstrates higher strength at cryo temperatures than at room temperatures
- Effect of ductile to brittle transition and beta Sn to alpha Sn transformation is still under investigation

In-Situ Micromechanical testing at **elevated** **and** **cryo** **temperatures** (Ongoing work)

Four compositions

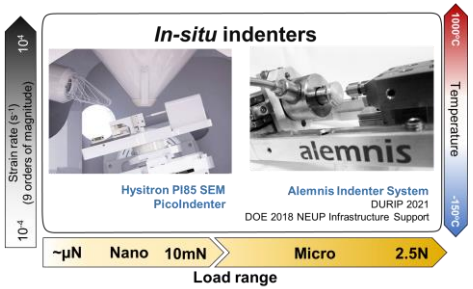


1) Pure Sn
2) 40Pb-60Sn
3) 50Pb-60Sn
4) 95.5Sn-3.8Ag-0.7Cu

Testing temperature (°C)	Homologues Temperature (T_{test}/T_m)					
	40Pb-60Sn/ 50Pb-50Sn	95.5Sn-3.8Ag-0.7Cu	Sn	Pb	Ag	Cu
25	0.65	0.61	0.59	0.50	0.24	0.22
50	0.71	0.66	0.64	0.54	0.26	0.24
75	0.76	0.71	0.69	0.58	0.28	0.26
100	0.82	0.76	0.74	0.62	0.30	0.27
125	0.87	0.81	0.79	0.66	0.32	0.29
0	0.60	0.56	0.54	0.45	0.22	0.20
-40	0.51	0.47	0.46	0.39	0.19	0.17
-85	0.41	0.38	0.37	0.31	0.15	0.14
-145	0.28	0.26	0.25	0.21	0.10	0.09



In-Situ Micromechanical testing at **elevated** **and** **cryo** **temperatures** (Ongoing work)



Four compositions

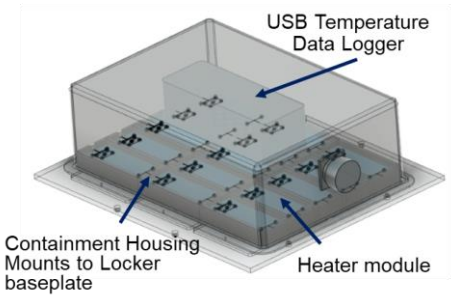
1) Pure Sn
2) 40Pb-60Sn
3) 50Pb-60Sn
4) 95.5Sn-3.8Ag-0.7Cu

Testing temperature (°C)	Homologues Temperature (T_{test}/T_m)					
	40Pb-60Sn/ 50Pb-50Sn	95.5Sn-3.8Ag-0.7Cu	Sn	Pb	Ag	Cu
25	0.65	0.61	0.59	0.50	0.24	0.22
50	0.71	0.66	0.64	0.54	0.26	0.24
75	0.76	0.71	0.69	0.58	0.28	0.26
100	0.82	0.76	0.74	0.62	0.30	0.27
125	0.87	0.81	0.79	0.66	0.32	0.29
0	0.60	0.56	0.54	0.45	0.22	0.20
-40	0.51	0.47	0.46	0.39	0.19	0.17
-85	0.41	0.38	0.37	0.31	0.15	0.14
-145	0.28	0.26	0.25	0.21	0.10	0.09



ISS Flight 2 (Summer 2024) soldering experiment: New microgravity experiments with better control of parameters

a) Science housing

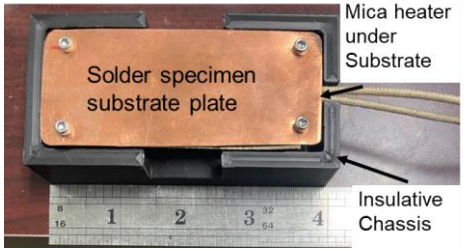


b) Microgravity Research for Versatile Investigations (MaRVIn) system



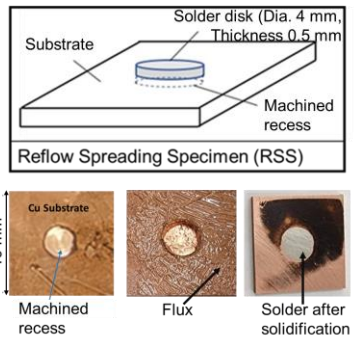
Scott Gilley, Tec-Masters, Inc, Huntsville
Jacob Scoccimerra, Nanorack, Houston

c) Substrate heater module

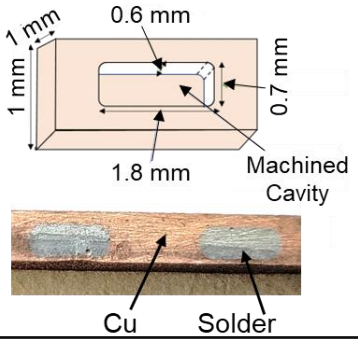


Each Solder specimen substrate plate will contain 40 to 60 solder coupons

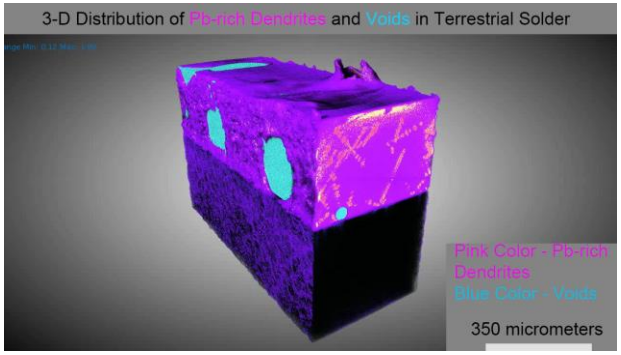
Large coupon



Small coupon



APS data: High-resolution tomography scans to characterize 3-D distribution of voids and Primary Pb-rich dendrites



Acknowledgments:

- NASA PSI (#80NSSC20K0223 – FY2020)
- NASA PSI (#80NSSC23K0279 – FY2023)
- NASA EPSCoR(#80NSSC21M0229)
- Dr. Louise L Littles (NASA Marshall Space Flight Center)
- Dr. Richard Grugel (NASA Marshall Space Flight Center)
- Dr. Binayak Panda (NASA Marshall Space Flight Center)

ISS Implementation partners:

- Scott Gilley, Tec-Masters, Inc, Huntsville, Al
- Jacob Scoccimerra, Nanorack, Houston, TX

UG Students

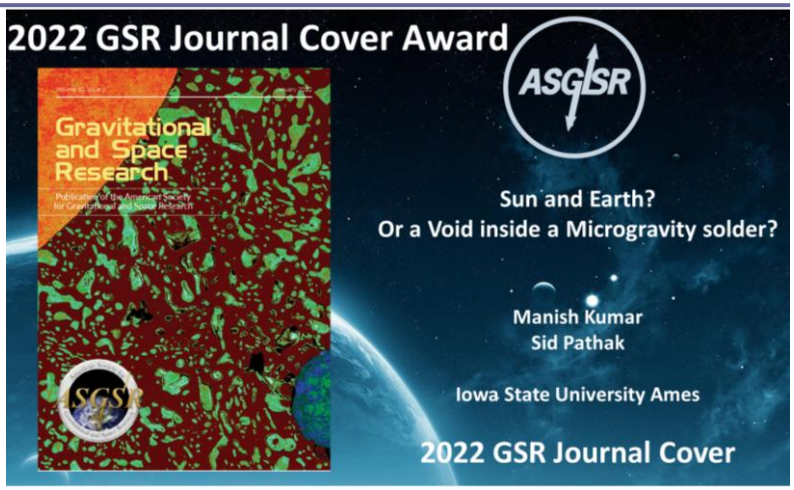
- Soren Hellyer
- Noah Madison
- Caleb Ogden
- Joshua Kamp

Instruments and facilities

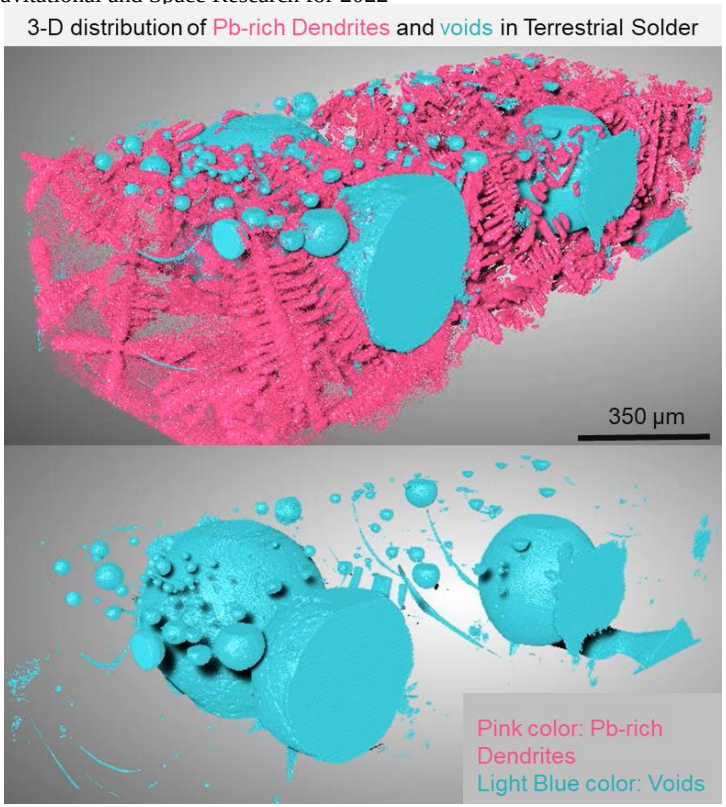
- Advanced Photon Source, Argonne National Lab, Lemont, IL
  - Dr. Alan Kastengren

Nano-mechanical testing facilities in our lab





Ph.D. work featured as the cover of the ASGSR's Open-Access journal Gravitational and Space Research for 2022



Studying the Effects of Aging on the Structure and Properties of Off-Eutectic Pb-Sn Solder Joints for In-Space Applications

Soren Hellyer, Caleb Ogden, Manish Kumar, Dr. Sid Pathak

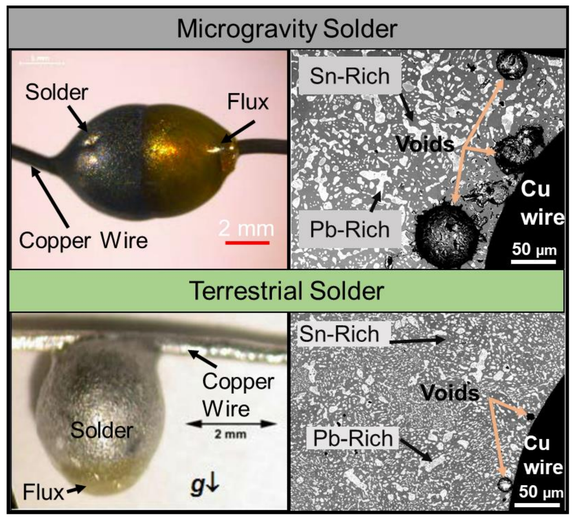
Primary Objective: Understand the effects of room temperature aging on the structure and properties of off-eutectic 50wt%Pb-50wt%Sn solder using scanning electron microscopy and nanoindentation.

Motivation: In-Space Electronics Repair and Metal Joining for Space Missions

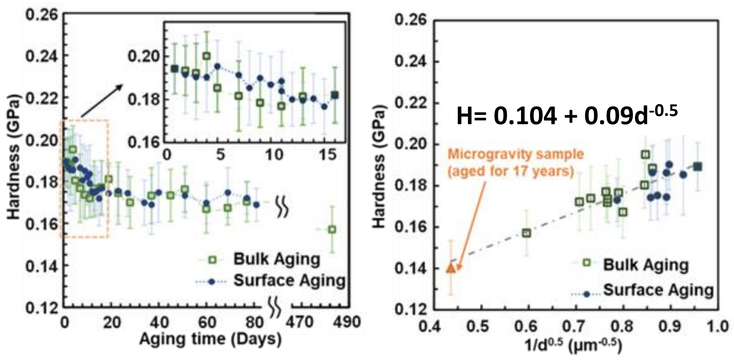
- Soldering is an essential metal joining technique for future extraterrestrial missions. *(NASA Mars Rover projects, manned missions to the moon and Mars.)*
- Can be used for fabrication and repair on missions.
- Future experiments will be studying the effects of microgravity on off-eutectic solder.

In-Space Soldering Investigation (ISSI)

- Understand the effects of microgravity on microstructure and void formation on sample solidification by analyzing samples from the ISSI ('03 - '04).

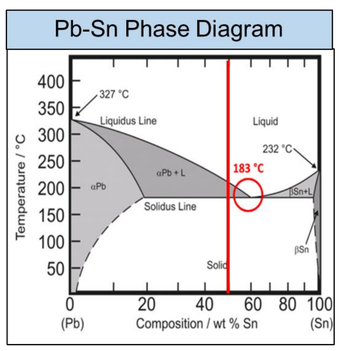


- Previous work done in the group shows that *eutectic* solder undergoes rapid aging at room temperatures.



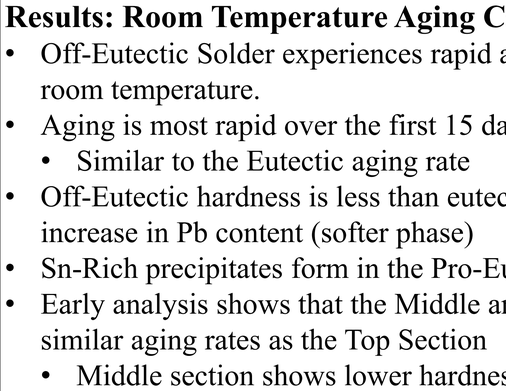
ISS Flight Opportunity: New Soldering Experiments aboard Space Station

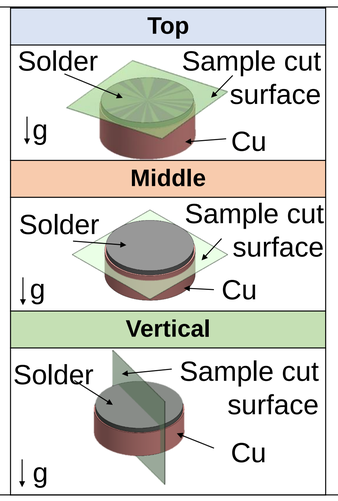
- Microgravity tests will be repeated about upcoming ISS flights
  - 40wt%Pb-60wt%Sn (Near-Eutectic)
  - **50wt%Pb-50wt%Sn (Off-Eutectic)**
  - 95.5wt%Sn-3.8wt%Ag-0.7wt%Cu (Lead Free)
- Samples may take months to return to Earth for examination.
- Grain coarsening occurs in metals when above 40% of their melting temperature (T_m)
- Room temperature is 60% of T_m for Off-Eutectic Solder
- Analysis of the samples will require understanding which changes in properties are due to microgravity and which changes are due to sample aging.



Methods

- Solder disks were solidified in Cu substrate
- 5 regions were studied
  - Sectioning to examine effect of gravity on aging
  - Testing on center and edge to test the effect of the substrate.



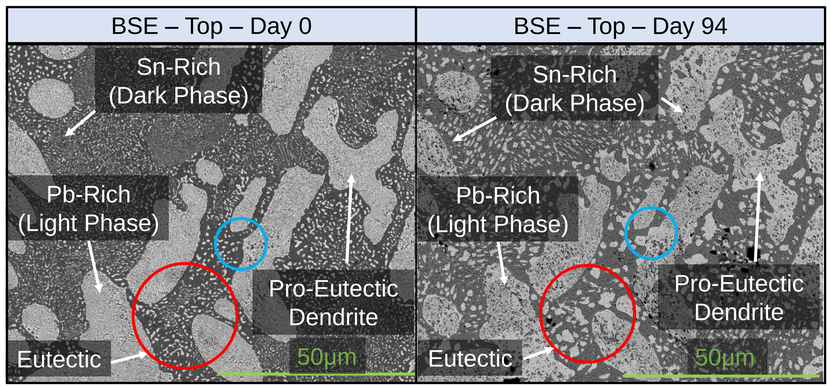


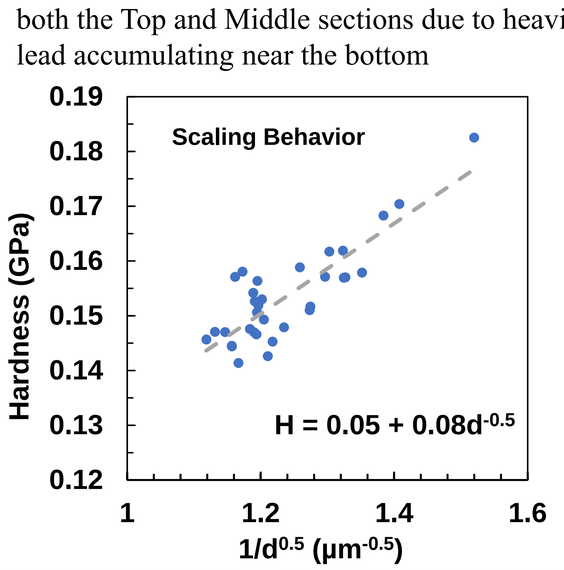
	Task	Description
A	Sample mounting and mechanical polishing	1. Sample mounting using epoxy 2. Sectioning each sample 3. Mechanical polishing up to 0.05µm surface finish
B	Microstructural characterization	I. Optical microscope imaging II. SEM imaging using FEI Inspect F50 SEM i. Secondary electron imaging ii. Backscatter electron imaging
C	Nanomechanical testing	Hardness and modulus measurement using Hysitron Triboindenter TI950 for local (Single phase) and bulk (multi-phase) response

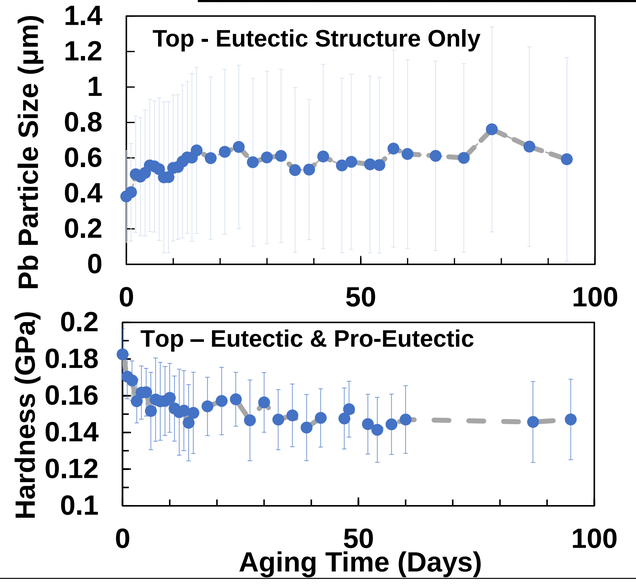
Day 1	Day 2 – 16 (study every day)	Day 19 – 61 (study every 3rd day)	Day 66 – 96 (study every 6th day)
Tasks A, B, C	Tasks B, C	Tasks B, C	Tasks B, C

Results: Room Temperature Aging Cause Grain Coarsening and Reduces Hardness (Strength)

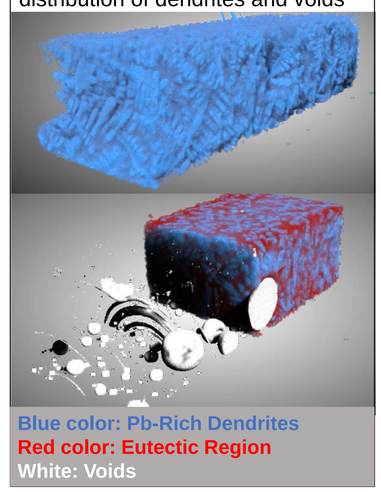
- Off-Eutectic Solder experiences rapid aging and grain growth at room temperature.
- Aging is most rapid over the first 15 days after solidification.
  - Similar to the Eutectic aging rate
- Off-Eutectic hardness is less than eutectic hardness due to an increase in Pb content (softer phase)
- Sn-Rich precipitates form in the Pro-Eutectic Pb-Rich Dendrites.
- Early analysis shows that the Middle and Vertical sections show similar aging rates as the Top Section
  - Middle section shows lower hardness than the Top section
  - Vertical section shows lower hardness than both the Top and Middle sections due to heavier lead accumulating near the bottom







Non-destructive tomography performed at the APS showing the distribution of dendrites and voids



Future ISS Flight experiments (Summer 2024):

Goal is to understand the causes of void formation and identify potential mitigation strategies

- Precise control over thermal profile such as temperature, heating and cooling rates, and max hold temperature
- Experiments planned for all three compositions

a) Science housing b) MaRVIn system c) Substrate heater module



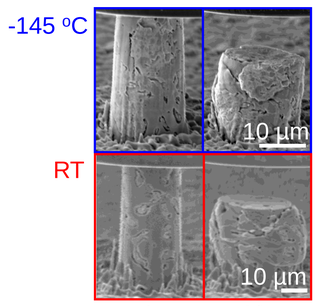
Science housing contains 12 solder specimen modules

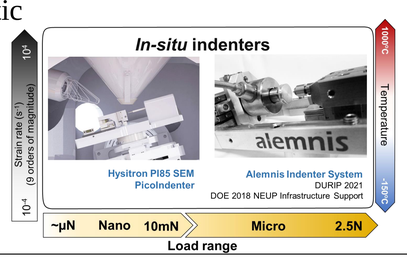
Each Solder specimen substrate plate (75 mm x 38 mm) will contain 40 to 60 solder coupons.

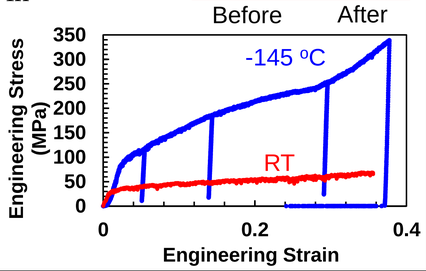
Effect of Temperature on Grain Coarsening & Hardness

Further work is planned on extreme temperature conditions

- The ISS experiences temperatures from +121°C down to -150°C
- Elevated temperatures lead to accelerated aging
- Sn phase change below +13°C
  - 27% volume increase, β (DC) → α (BCT)
- Ductile to Brittle Transition below -80°C in Eutectic







Acknowledgements:
NASA Physical Sciences Informatics (PSI) Grant **(#80NSSC23K0279 – FY2023)**
NASA EPSCoR **(#80NSSC21M0229)**
NASA Iowa Space Grant Consortium **(#80NSSC20M0107)**
Undergraduate Students
Joshua Kamp & Noah Madison

References

R. Grugel et al., "Final Research Report. In-Space Soldering Investigation (ISSI)," 2006